

TAYLOR'S PROGRAMMES MAY 2026 SEMESTER

ITS68404 – Data Visualization

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Executive Summary

The Kalimati market at Kathmandu is the biggest wholesale produce market in Nepal supplying 60–70% of the vegetable demand in Kathmandu Valley, which gets an estimated 700–800 tonnes of vegetables each day, on an average. An interruption to that chain of supply has an immediate and tremendous impact, as shown by the drop in daily arrivals during a disruption in the chain of supply, triggered by the rain, in late 2025, to less than half the usual amount, and the increase in the wholesale prices of green vegetables up to 225 percent in a single day. This project is designed to study the influence of temperature, precipitation, humidity and wind speed on climate variability on the price movements of wholesale vegetables commodities in Kalimati with a merged data set of 2013-2019 for 128 commodities.

The problem is not that this relationship is unrecognized. The government-backed Kalimati Fruits and Vegetable Market Development Board (KFVMDB), in charge of the market since 1987, has already been monitoring and publishing a wholesale price index for the market daily. What is missing is any tool that lets the Board, or the agencies that rely on its data, see that price index alongside the weather conditions driving it. Existing sources are either static and siloed from weather data, too aggregated to be commodity-specific, or contain no visualization at all leaving a genuine, well-documented problem without an interactive way to explore it.

This report presents a dashboard designed for the KFVMDB and allied agencies such as the Department of Agriculture, to support three concrete decisions: identifying which commodities are most weather-exposed in a given season so alternate sourcing from supply districts such as Dhading, Chitwan and Kavrepalanchok can be arranged in advance; timing public price advisories ahead of anticipated monsoon-driven disruptions rather than after prices have already spiked; and informing longer-term buffer-stock and storage planning by season.

1. Project Proposal

1.1 Chosen Track & Nepali Context

This project was proposed in the following data-for-good stream: Data Visualization to solve a real-life public interest problem in Nepal – Kalimati Fruits and Vegetables Market in Kathmandu and linking the price volatility of vegetables in the market with the effects of climate variability.

Though agriculture in Nepal remains an important sector of the economy and livelihoods (around sixty per cent of households in Nepal are engaged in agricultural activities) and is rain-fed and highly vulnerable to monsoons. The vegetable market of Kalimati is the biggest wholesale produce market in the country providing vegetables for the retail markets and households in the Kathmandu Valley. Vegetables are perishable commodities, affected by the immediate loss of standing crop, and one big rain with rain enough in days can wipe out the standing crop or deplete supplies. As per the historical data of Nepal Rastra Bank (NRB) (Section 4), fruits and vegetables are one of the most volatile sub-components of the CRP and that has been reflected in the recent reports of the food inflation entering into double digits during the previous pre-monsoon supply disruption in 2024 and the consumer inflation reaching a 16-month peak of 5.04 percent by mid-2026.

Even with this known volatility, there is no public interactive tool to explore the price–weather relationship at the commodity level that a Nepali stakeholder could use to do so. This project will create an interactive Tableau dashboard that will combine Kalimati price data over the last seven years (2013-2019) with climate data.

1.2 Chosen Datasets & their sources.

Dataset	Source & Coverage
Kalimati Tarkari Dataset (Fruits & Vegetables Price)	The data on fruits and vegetables prices is from Kalimati Tarkari Dataset, and it is provided by Kaggle / Open Data Nepal, licensed CC BY 4.0. Wholesale market prices of minimum, maximum and average prices of 136 commodities procured from Kalimati Fruits and Vegetables Market, Kathmandu on the daily basis. The biggest wholesale produce market in Nepal. Timeline: 16 June 2013 – 28 September 2023 Number of records: 293,641 records.

Nepal Daily Climate (1981–2019)	Data was obtained from Nepal Daily Climate (1981 – 2019) and Kaggle (Saimon Dahal) from the NASA Langley Research Center (LaRC) POWER Project. Daily temperature, precipitation, humidity, wind speed and wind pressure data were obtained for 93 weather-station sites across 62 out of 77 districts of Nepal. Timeline: 1 January 1981 – 31 December 2019 Number of records: 883,128 records.
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The two sets share a common Date field, and the datasets will be merged and cleaned (deduplication, type correction, district-name standardization, etc.) as a basis for the dashboard.

1.3 Target Audience

The dashboard is not designed for public use, but rather for a market-facing audience, with a focus on decision makers:

- The primary Kalimati Fruits and Vegetable Market Development Board (KBV) could benefit from having information on the seasonal price risk patterns to time public price advice and communicate with the supplying districts in advance to the expected weather disturbance.
- Vendors and traders might benefit from reacting to commodity price volatility and seasonality to avoid exposures to commodity price peaks and valleys.
- Policy facing analysts such as those in the agriculture ministry or those near central banks could be used the climate–price linkage evidence to guide decisions on subsidy timing or import-policy decisions for the most weather-sensitive commodities.
- The dashboard was designed for consumers and journalists (secondary) to help understand and predict price changes that are only reported afterwards.

Nepal is a geographically diverse and a landlocked country where agriculture continues to play a major role in people's livelihood and in the economy of the country. In Nepal, agricultural dependent households account for around 60 per cent, of which majority are rain-fed agriculture and are quite vulnerable to monsoon timings. This is a larger scale picture, where the Kathmandu Valley is a strange place as a vegetable producer, with little vegetable production in the Valley itself or available locally for use; it uses almost exclusively vegetables imported from the surrounding districts Dhading, Chitwan, Makwanpur, Kavrepalanchok and the Terai.

That dependency is directed almost exclusively to one institution. In 1987, Kalimati Fruits and Vegetable Market Development Board was formed as the first organized terminal wholesale market in Nepal with the support of Government and United Nations Capital Development Fund. It is now the biggest wholesale produce market in the country having several hundred wholesale traders and it fulfils the demand of 60-70 per cent of the vegetable requirement in Kathmandu valley within a span of one day with 700-800 tons of vegetable produce. This market is a key supply route for Kathmandu, and blockages in the roads and districts to it make themselves immediately apparent, in the price in Kathmandu.

This sensitivity isn't abstract. The last few cases have happened firsthand: If the supplies were interrupted in the end of 2025 because of rain, the number of green vegetables decreased to 387 tons, while the usual amount is 700 tons and the wholesale prices of green vegetables rose up to 225 per cent one day. In a similar incident of 2024, after the monsoons, the heavy rainfall resulted in extensive damage to the crops and affected the transport routes and the year-on-year inflation rate of commodities arrived at 9.84 percent, while the headline inflation rate was 3.85 percent, which means vegetables' share of weather-related inflationary pressures was disproportionately high. After several months of disruption to the road system into the Capital due to flash floods in late September, vegetable inflation had risen even further in December 2024, to 43.05 percent year-on-year.

The availability of these data is indeed their key characteristic; the data on prices which such decisions would be based upon are already collected and published by KFMDB itself through its own daily wholesale price index, making this problem, and indeed everything else, tractable. This project touches directly on that price data and its visibility due to the weather; there is no way to analyse that price data along with the weather that makes it so plainly visible. By combining 7 years of Kalimati wholesale price data and climate data (2013 to 2019), we create an interactive dashboard that allows the Board and allied agencies to gain insights into the weather-price relationship of an individual commodity, season or year as opposed to relying on the static reports, aggregated indices or after-the-fact news coverage currently providing a substitute for this analysis. The focus on the audience, from a general public audience to a specific one of a named institutional stakeholder who may have some interest in acting on the

question at hand, as well as narrowing the focus of the analysis, from commodity level patterns to patterns at the district level, was refined throughout the project by ongoing academic consultation with the module leader Anmol Adhikari, whose guidance continued to focus the analysis away from a broad descriptive account of food inflation in Nepal and into a narrower focus that was relevant to the decision. The remainder of this report discusses how we conducted our research, how we were able to integrate the data with our methodology, how we built our dashboard, and a critical analysis of the product of our research.

2. Background Research & Literature Review

To comment on the fluctuation of the prices of vegetables in the vegetable market of Kalimati it is necessary to understand the relationship between climate and agriculture in Nepal in general. It doesn't only list these 7 sources, in 3 categories, 2 peer-reviewed academic studies, 1 official report from a central bank, 4 current news reports, each reporting one of the various, dated, price events; it combines them all so that what it does and what it can't be compared. The author does not just judge the substantive findings of each source but also the methodological reliability and scope of the source for determining the weight to be accorded to it as a source for substantiating the approach to be taken in this project.

2.1 Peer-Reviewed Sources

National food production and climate change. Neupane et al. (2022) used national agricultural and climate data to examine the district-year resolution scale relationship between precipitation trends with the production of cereals in Nepal's seventy-seven districts over 1990–2018 in *Frontiers in Sustainable Food Systems*. They determined an annual production of the main crops was directly proportional to the changes in average annual precipitation, and that districts with more floods and heavy rainfall events had higher malnutrition rates during the same time span. The authors also note the correlation between vegetable crop losses due to frosts and the unpredictable nature of the rainfall and pest pressures, with one district reporting a nearly 50% loss of potato production one year because of frosts. The best features of the study methodologically are that it covers a length of time, 30 years of production records in seventy-seven districts, and that no shorter or more local study would give the national level conclusions which it has reached and justify as such. The district-year aggregation is the project's main limitation, however, as it can only illustrate the trend of the yearly totals relative to the total

rainfall for the year, at a resolution much cruder than the daily, commodity-level resolution needed by a wholesale-price dashboard. They're interested, that's, in production and nutritional outcomes not in production to the nutritional, but at the district level, what occurs in the field, and then on the plate, not between, at the wholesale market.

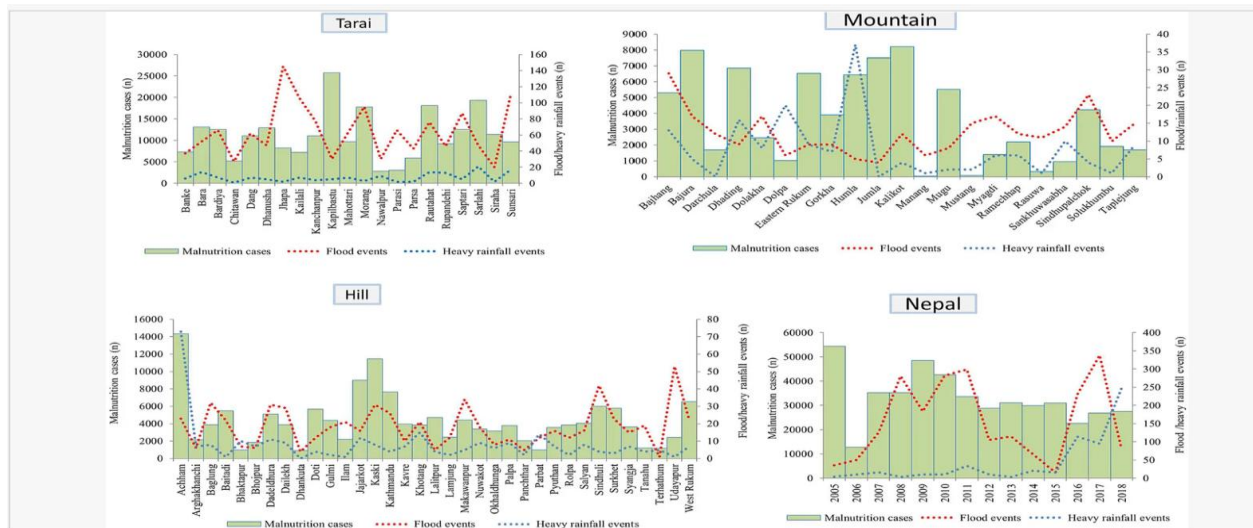


Fig1: Ecological and district-wise distribution of malnutrition cases, flood, and rainfall events from 2005 to 2018 in Nepal

Impacts of climate change on market systems in agriculture. Taking the analysis from the field to the market, Barrueto et al. (2017) look at Sustainability. They did not study yield per se, but in the context of the specific impact of climate change on processing, storage and sales, they carried out qualitative fieldwork in fourteen districts and various sub-sectors of agriculture in Nepal, including vegetables. They have forecasted an increase in monsoon and post-monsoon rains and a decrease in winter rains and traders have already said that there has been an increase in the number of windstorms that are forcing them to miss the harvesting and fruiting periods. One of the key strengths of this research is the market-systems approach it provides to discuss the price and supply chain disruption as relevant climate impacts, alongside yield loss. It is also, of course, a methodological middle-ground recognized and accepted from the outset: It is a qualitative study of districts from interviews and observation in the field and thus cannot provide the kind of statistical generalizability that a district-year panel study can. The results, however, are more likely to be interpreted as evidence that the mechanism exists and cannot be interpreted as an estimate of the magnitude of the mechanism, a place where the continuous, quantitative dataset on which this project builds is well suited to provide such interpretation. Barrueto et al. (2017) take this process a step further to the farm gate and show the same chain of causation for storage losses and compromised sales timing, the latter of which is directly

linked to wholesale prices, in this project. When combined, these two studies support the notion that the climate affects the agriculture of the Nepalese people, and that the effects are not only felt on the farm, but also in the marketplace which is unique to the dataset used in this project.

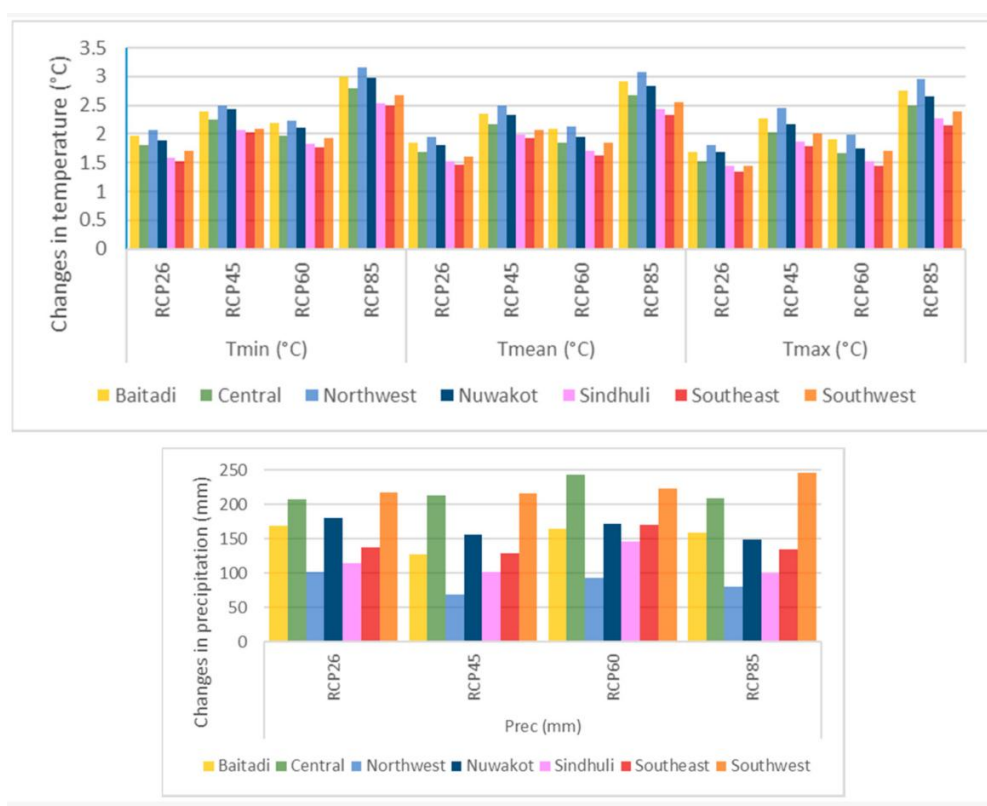


Fig2: Depicted changes of climate and precipitation

2.2 Official Government Report

Historical price volatility. In contrast, Nepal Rastra Bank's special publication on inflation for 2007 provides a different perspective as it provides explanation of Nepalese inflation mostly from the perspective of price transmission from India and the historical data is directly applicable here. During this period of 30 years (1976-2006), the score of the "Fruits and Vegetables" sub-component of the Food and Beverage group of the national CPI basket was volatile at 1.02, which is higher than the average score (0.79) of the national CPI basket. (Nepal Rastra Bank, 2007). It is a national, Government produced index and as such is highly reliable for the specific claim it backs: Vegetables are structurally more volatile than food prices generally. This volatility is exactly the kind of detail the data set for this project is intended to offer, however; the CPI sub-component combines dozens of individual fruits and vegetables and into a monthly number, so it is hard to pinpoint any of this volatility to a specific commodity, market or weather event. This provides an important point of contrast to the studies

of Neupane et al. (2022) and Barrueto et al. (2017): the price differences in vegetables are not only a climate issue; they also have a long history of being monetary issues and, as the earlier analyses show, they provide additional pressure on the market from climate change. This project does not discount the role that other variables play in the price volatility of the Kalimati, it is only an underestimated, measurable one.

	Items	Mean	Std. Dev.	Volatility*
	Overall Inflation	8.23	4.61	0.56
	Food and Beverage	8.17	6.47	0.79
1	Cereal Products	7.43	10.22	1.38
	Rice	7.48	10.77	1.44
2	Pulses	9.56	11.41	1.20
3	Fruits and Vegetables	9.24	9.44	1.02
4	Spices	9.09	15.02	1.65
5	Meat, Fish and Eggs	9.04	4.76	0.53
6	Milk and Milk Products	8.55	5.45	0.64
7	Edible Oil and Ghee	8.20	14.73	1.80
8	Sugar and Sugar Products	7.68	14.52	1.89
9	Beverages	8.35	7.03	0.84
10	Restaurant Meals	10.33	8.37	0.81
	Non-food and Services	8.39	2.85	0.34
1	Cloth, Clothes and Sewing	6.84	3.74	0.55
	Clothes	6.08	4.62	0.76
	Sewing	7.10	3.96	0.56
2	Footwear	6.08	4.22	0.69
3	Housing	9.98	3.63	0.36
	Fuel, Light and Water	11.64	6.22	0.53
4	Transport and Communication	8.52	4.76	0.56
5	Medical and Personal Care	7.15	3.90	0.55
6	Education, reading and Recreation	8.77	5.47	0.62
7	Tobacco and Cigarettes	6.85	4.82	0.70

Fig3: Table of inflation performance

2.3 Contemporary News Evidence

The other 4 sources are used individually, record separate, dated events with their own set of numbers and in combination they significantly bolster the evidence base above, the 2 academic studies and the one government report.

The monsoon season is a major factor in vegetable price fluctuations, according to the report, "Monsoon pushes up vegetable prices," in The Kathmandu Post (July 2024). According to this report, the vegetable price index shot up by 23.11 percent only during the pre-monsoon period

(mid-May 2024) as the winter drought was compounded by early monsoon heavy rainfall, causing floods and landslides, as revealed by Nepal Rastra Bank's own macro-economic data. The information officer of Kalimati market disclosed that it reduced from 700 tons to 500 tons of its supply after the onset of rain. The prices of individual commodities are not fixed to a single average and showed significant variations in commodity prices within a month with onion prices increasing 47.09 percent to Rs 97.67 per kg, cauliflower rising 72.73 percent to Rs 95 and okra up 154.63 percent to Rs 55.

The data comes from the information "Vegetable prices soar above Rs100 per kg" of The Kathmandu Post (October 2024). About 5-6 months later, the same market was still short of supply and receiving 500-507 tons per day, which was 200-207 tons less than the usual amount of 700-800 tons. The article cited the data from the Nepal Rastra Bank (NRB) that overall consumer price inflation was at 3.85 percent in mid-September this year while the inflation rate of vegetables was 9.84 percent. The wholesale price of tomato had reached Rs 125 per kg while retail price was in the range of Rs 200 to 210. One of the most salient facts in the literature reviewed here is that the vegetables price index is more responsive to a supply shock than the rest of the price index.

The price of food is again rising, January 2025, Kathmandu Post. In mid-December 2024, the overall food inflation rate stood at 9.99 per cent and the headline consumer price inflation rate was 6.05 per cent year-on-year, while the vegetable inflation rate continued to increase to 43.05 per cent year-on-year. This is due to "extremely heavy precipitation" during the days 26-28 September 2024, which caused flash floods that impacted road facilities to Kathmandu for "several months", the article says. Importantly, the economists mentioned in this article also highlight other non-climate factors such as depreciation of the Nepali rupee in comparison with USD, import ban of vegetable staples from India and insufficient monitoring of the markets to show the nuance already suggested in Nepal Rastra Bank (2007) report: climate is a major and measurable cause of volatility in vegetable prices, but not the only one in operation at any given time.

According to "Vegetable prices in Kathmandu rise sharply after rains hit supply routes," published in The Kathmandu Post in November 2025, the prices of vegetables in Kathmandu are going out of hand due to the rains that hit the supply routes. The most recent evidence is that a year later the daily arrivals at Kalimati were reduced to 387 tons (down from the usual 700 tons) due to heavy rains and wholesale prices of green vegetables rose up to 225 percent

in one day. The similarities between these five documented events, which all occurred in a similar way (rainfall — disruption — shortages of supplies — price increase), suggest that the structure of the weather response of the Kalimati market is not an isolated occurrence, but rather a structural component.

2.4 Synthesis Across All Seven Sources

These sources offer a coherent set of evidence rather than a set of individual claims. Neupane et al., 2022 and Barrueto et al., 2017 proposed a mechanism at a broad multi-district level – climate affects production and functioning in both the production and market. The report by Nepal Rastra Bank (2007) is a very important note of caution – the volatility of the price of the vegetables is not solely attributed to climate change; it has several non-climate causes as well. The four Kathmandu Post reports then, however, give the academic and government sources what they lack: market-specific, dated examples to test the general mechanism against actual numbers from the same market, from which this project's data was extracted. Seven sources were used in this project as none of the sources in this review gave all three properties: generality, mechanism and market-specific evidence.

2.5 Research Gap

These seven are available and can be organized on a continuum of spatial and temporal scales, and can be displayed on the same scale, to allow visualization of the gap at an exact scale. At the district-year time resolution level Neupane et al. (2022) are not market specific and have broad coverage (77 districts). Barrueto et al. (2017) conduct their analysis qualitatively for fourteen districts (richer mechanism, but not quantitative, and not specific to a single market's transaction records). The report by Nepal Rastra Bank (2007) is based on the national components of CPI, monthly and long-time (30 years) period but it is too aggregated to fix a particular commodity, and it doesn't have a direct connection with daily weather. The four Kathmandu Post articles are on the other side: They are specific to Kalimati and specific to individual commodities and only last a week or a month, and there is no continuity between one and the other.

The data set used for this project includes only one market, in commodity level, daily time-step, multiple years of data and directly incorporates daily weather data for the same time and

place (all of which are not included in the data sets reviewed here). This is not an absence of degree but absence of kind; none of the existing studies could, even in principle, answer the specific question this project asks, namely, not "whether or not climate affects Nepali agriculture" (which is well established) but rather "what is the amount of price fluctuation of a given commodity that can be attributed to a given day's weather at Kalimati specifically. This project offers the following five dimensions which are not covered in the literature presented above, or which are covered in only one or two dimensions:

- spatial resolution (a single market instead of seventy-seven districts and/or a national aggregate),
- temporal resolution (daily instead of yearly or monthly),
- aggregation level (an individual commodity instead of a CPI sub-component or a whole agricultural sector),
- the absence (established separately below) of any decision-support visualization built on top of the data even at the level of spatial and temporal resolution described above),
- the absence in the data, even at the level of spatial and temporal resolution described above, of any decision-support visualization built on top of the data. It's not any single one of these that makes this project special, it's all of them together.

It's also important to note that this project's own data spans 2013-2019, and that the most dramatic and dramatic events of the last two years (2024-2025) make up the top five news events. Literature and news evidence thus confirms the reality and contemporality of the mechanism that this project explores, but the dataset itself has the ability to demonstrate the presence of the mechanism only for the period 2013-2019.

Dimension	Existing Literature	This Project's Contribution
Spatial Resolution	District-level (Neupane et al., 77 districts) to national CPI aggregate (NRB); news sources anchored to Kalimati but only as isolated, dated snapshots	A single market (Kalimati), observed continuously rather than snapshot-by-snapshot
Temporal Resolution	Annual, district-year data (Neupane); undated qualitative fieldwork (Barrueto); monthly over thirty years (NRB); a single week or month per event (news sources)	Daily time-step, continuous across 2013–2019
Aggregation Level	Whole-crop or district totals (Neupane); qualitative sub-sector themes (Barrueto); a CPI sub-component bundling dozens of items (NRB)	Individual commodity, across 128 tracked items

Data Integration	No source merges weather observations with wholesale price records at matching granularity	Weather and price records merged at daily, commodity level for the same period
Decision-Support Capability	Descriptive indices, static reports, or narrative journalism; no interactive tool in any source	Interactive dashboard supporting three specific KFVMDB and allied-agency decisions

Table 1: Comparative positioning of existing literature against this project's contribution, summarizing the discussion above

2.6 Visualization Gap

Visualization gap is like the research gap but different in that it concerns communicating the relationship, rather than if it exists or not. The gap is specified by three points. First, the infrastructure that this project builds on, the KFVMDB, provides it: the KFVMDB publishes the daily index for wholesale prices, on which this project is based, and the data is public, but they don't offer any chart, graph, or interactive tool to explore the data, let alone compare it with the weather — the Kalimati Tarkari dataset is created using exactly the same data that the Kalimati Tarkari is based on, and they make that data public, but they don't provide any chart, graph, or interactive tool to explore it, let alone to explore it in conjunction with the weather. Second, none of the three visualizations covered in the next section solves the complete problem; Trading Economics offers interactivity but lacks a granular level of commodity representation; OnlineKhabar allows for a readable individual chart but lacks the visual encoding of related variables; and The Kathmandu Post's own reporting offers timely commentary without any visual encoding at all. It is not possible to have a price series and a weather series in the same interactive frame for any of the three. Thirdly, this is not just a desktop cost of this gap, but an operational cost – the very information officers of the KFVMDB will quote when the news comes out, about "look at all the rains and road closures". This is a qualitative, not a dashboard/real-time based, real time gap cost. Thirdly, however, it is an operating cost: when the price increases described above occur, the KFVMDB's own information officers will be cited above as saying "look at all the rains and road closures". This project's dashboard is intended to fill that final mile gap, not to uncover something new about how climate impacts Kalimati prices, but rather to make that well-known relationship accessible, in a “nearly real time” fashion, to those who are compelled to figure it out after the event, using memory and news articles. In practice, closing the last-mile gap is an iterative design problem, and was directly influenced by consultation with the module leader, Anmol

Adhikari, who asked each of the candidates' charts whether it answered a challenge the KFVMDb, or an allied agency, actually had to make, as opposed to just providing a collection of data for the sake of it – which allows the scope of the dashboard to remain narrow enough to be usable without all the unfocussed complexity that the existing alternatives below raise the spectre of. The same consultation also acted as a template for the same core visualization concepts to be applied throughout the design process: Gestalt's grouping laws, Tufte's data-ink ratio, and Anscombe's (1973) insights into the potential for summary statistics to be similar looking but represent very different underlying relationships.

3. Critique of Existing Visualizations on the Topic

The five fundamental visualization principles considered in the evaluation of all three visualizations are Tufte's data-ink ratio principle (1983), which emphasizes the importance of using a chart to maximize the amount of ink used to convey data and not decoration; the Gestalt principles (1960) that consider how a viewer will visually group related data; principles of chart type – data type (Cleveland and McGill 1984) that consider the best type of chart for a given type of data; and the overall accessibility of the chart to the intended audience, including the cognitive load imposed on a non-technical viewer. Every critique ends with the explicit design answer this project's dashboard embraces in response.

Trading Economics — Nepal Food Inflation (tradingeconomics.com/nepal/food-inflation).

The design of this interactive line chart has reasonable data-ink ratio – it is minimal, the hover tooltips replace static labels, no unnecessary ornamentation or gridlines, and the design is suitable for the chart-type of a single continuous variable. It's not a bad performer, it is a bad viewer, it only presents an aggregated food inflation index, there are no commodity-level series, there are no weather variables and there is no filtering mechanism provided for a viewer to home in on a specific product or season. A viewer can see that inflation is rising and it is unlikely that a viewer will ask, "whose or what is inflation going up due to?" The chart is accessible to a public who want to know that inflation is rising, but not this project's audience who wants to know, "whose or what is inflation going up due to?". This is the sound choice of encoding, as one line represents the value by its position on a common scale, the best visual channel at which to perceive the value, in Cleveland and McGill's (1984) terms, a matter of scope and audience fit, not of appropriateness of the chart type. However, the project's relevance in real life is that it has been designed to answer the question of whether a price move

has occurred, but not who or what is its driver, whether that be the commodity, district or weather - the information vacuum that KFVMDDB is expected to address for its target audience.

Whereas the price could drill down on an aggregate line on Trading Economics, our dashboard offers the user a commodity-level filter and lets him/her drill down into any of the 128 commodities we track and view the price history of that commodity alone.

OnlineKhabar "The Nepali economy explained in 10 graphs" (english.onlinekhabar.com/nepali-economy-explained.html). It really does the work of the Nepal Rastra Bank with the loaded bulletin but breaks it down into 10 separate but readable graphs; it's pretty good at choosing the chart type and labelling the individual graphs. Its problem is that it violates Gestalt's proximity principle, that is, that the ten graphs are static images without given axes and that no one reader can see if two variables (such as inflation and remittances) vary together, since they are never put within the same visual frame or on a shared axis. This is also an accessibility issue: when reading, the reader would have to work out what the relationship is, whereas with the linked, shared axis view they would be able to see what it is directly without any working memory required. This is the kind of cognitive load that the linked, shared-axis view is supposed to save the reader, and it is this kind of cognitive load that a lot of people must expend when reading and comparing the shapes of the two graphs.

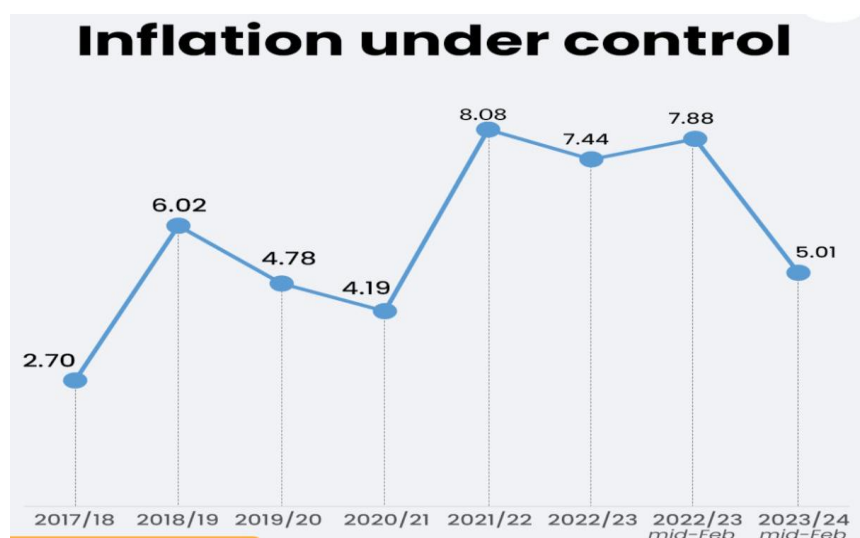


Fig4: Inflation rate in percentage.

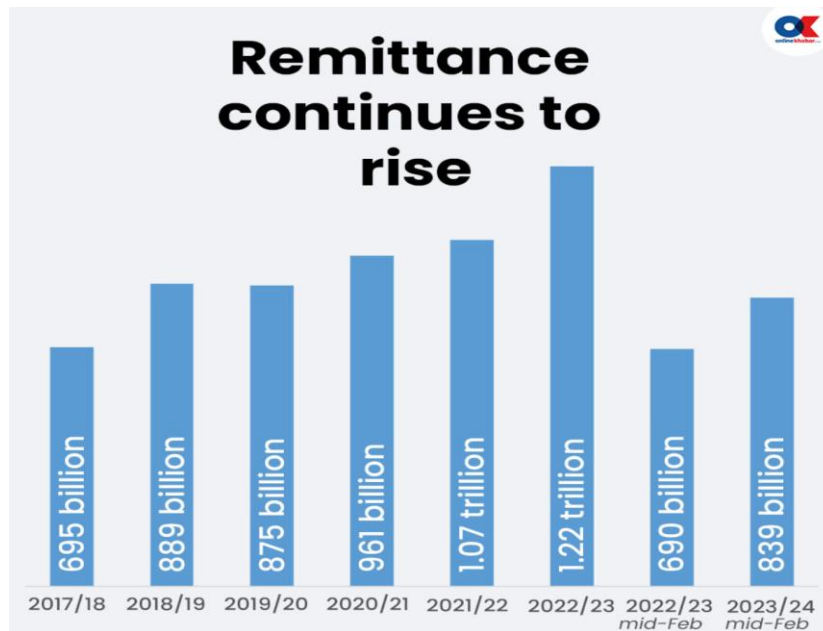


Fig5: Rise of remittance

Dashboard response: When the views are linked with common brushing, if I select a time range or a season in the price chart it will filter the weather chart on the same row. This brings cause and effect to the same chart so that the reader doesn't have to imagine what they would have seen on the weather chart if they had not selected the time range or season in the price chart.

The Kathmandu Post “Inflation cuts deeper as food and fuel costs strain households” (June 2026). It's well sourced, timely journalism, but there's no chart or graphic of any sort and the reader must retain the numbers in his head and follow them through several paragraphs of prose: “5.04 percent” and “4.63 percent. So, by Tufte's measure, this is the opposite of failure to chart-junk: Not that data-ink is cluttered by decoration, but that there's none at all, the information exists only as text, which is less accessible than the other two formats; the information is not accessible to a reader who wants to see a trend but rather wants to know more about a single data point. The practical implication for a decision-maker is not insignificant, however, because, when a KFVMDDB official gets this coverage, they must try to see a trend from the individual percentage numbers rather than see it first-hand – which this project's dashboard aims to help them with.

Every figure of price and weather on our dashboard is both visually and numerically coded (Gestalt similarity) by using a consistent colour scale for all the categories of the dashboard, thus allowing a reader to identify a category without reading a label every time.

Summary: The examples show that the stakeholder in Nepal in each case has an unmatched authority source and an interactive element, as well as a granularity of the commodity weather that is not a match. Trading Economics is interactive and well charted but overly aggregated; OnlineKhabar is accessible to any single chart of one of the related variables, but it lacks any of the gestalt proximity of the other variables some are spread out across chunks of static graphics; The Kathmandu Post's own current reporting has none of the data-ink whatsoever. These three failures, in turn, embody the four principles outlined above, and are the principles that the dashboard in this project aims to be built around: Granularity, Linked Proximity, Appropriate Chart Type and Visual Accessibility. Each source has its own structural limitations, as none will have been designed with the specific intention to support a commodity decision with a weather component and none of the existing commodity decision-support options a KFVMDB stakeholder has are currently being used to support such a decision.

Synthesis. The four strands of this criterion do not make four distinct cases but rather arrive at the same conclusion. The literature review substantiates that the effects of climate variability on the agriculture and markets of Nepal can be quantified and there is no existing data source available to monitor the effect on a single commodity on a single day at Kalimati in terms of scope and resolution. This research gap turns the statement into a five-dimensional (precise) conclusion on what this project's dataset can provide and that no tool exists which can turn that data into a view that is explorable and decision-relevant: These three strands of ongoing consultation with the module leader, Mr. Anmol Adhikari, were not mutually exclusive but instead reworked the project's framing, audience and design choices in relation to the single criterion of whether a particular KFVMDB or allied-agency decision would be better made as a result. It is this problem, which the literature has confirmed to be a problem through evidence, but cannot resolve to a desired resolution; it is this gap, which is well defined so that it can be closed, rather than described; it is these criteria, which have been tested by stakeholders and have defined what constitutes a useful design response, that justify the dashboard that is developed in this project and provide the terms on which the data integration, design decisions, and final evaluation in the following sections will be evaluated.

4. Stakeholder Consultation Insights

A series of project supervision sessions were held with the module leader, Mr. Anmol Adhikari, which focused on a structured stakeholder consultation to assess the appropriateness of the datasets used in the Kalimati wholesale price and district-level climate datasets for a decision-support dashboard.

Three themes were discussed during the consultation:

- the need to combine daily commodity price information with climate information,
- the methodological challenges of aggregating weather data at spatial scale,
- The value and potential use of the proposed visualizations for a named community of Nepali.

The feedback received in the consultation process revealed that the project scope was too general and descriptive, and it was recommended that the study was to be done based on the operational needs of the Kalimati Fruits and Vegetable Market Development Board (KFVMDDB). Further, after analysing the data, insight was gained on the influence of climate and meteorology on the prices of commodities in the city of Kathmandu as it is the centre while the others are producing the commodities.

The team therefore decided to focus the common data period (2013-2019) on price volatility as the most relevant indicator to monitor, and to re-engineer the dashboard to make each chart an answer to a specific stakeholder question, such as commodities vulnerable to weather shocks, seasonality of disruptions, visualization of district-level concentration of rainfall etc. The consultation also highlighted the three principles explicitly followed in the final dashboard layout and subsequent analytical worksheets: Gestalt grouping principles, data-ink ratio (Tufte's principles) and Anscombe's lessons on the limits of summary statistics.

5. Data Sources, Integration & Cleaning

5.1 Source Selection and Credibility

Two tabular data sources and a supplemental geographic extract are used in the dashboard. The price data (2013(June)-september-2023-cleaned.csv) consists of 293,641 observations across seven columns that represent the minimum, maximum and average price of commodities sold daily in Kalimati, wholesale market during the period June 16, 2013, to September 28, 2023. The climate dataset daily climate 2.csv is a daily station dataset, with 883,128 rows representing observations of temperature, precipitation, humidity, wind speed and related variables, broken down by district and geographic coordinate, and spanned from 1 January 1981 to 31 December 2019. They are both third party compilations (not raw institutional feeds) and have both been subjected to internal consistency checks during the pre-processing as such the credibility of sources is not based on provenance documentation provided with the files themselves, but rather on the internal consistency checks performed during pre-processing; this is acknowledged here and is not assumed away. A third data source, district-level climate extract, was also located away from the price-climate extract and only used for the geographic ("Rainfall by district") view. This was not a coincidence because if all the climate information at the station level were combined to a single national data point, the spatial variability that is at the heart of the meaning would be destroyed. This is a direct consequence of climate data needing to be provided at two levels of spatial aggregation for the two views and is discussed further as a limitation.

5.2 Cleaning and Type Correction

The structure and data types of these datasets were loaded and studied prior to any transformations. In both datasets, there was a date column, and, in both cases, it was a string – it's not actually a date column in Tableau. A duplicate check was performed on the aggregated climate table by date, and no duplicate rows were found, which is an assumed (not verified) result for all cleaning actions in accordance with the requirement to verify, not assume, a cleaning action.

5.3 Integration Strategy and the Aggregation Decision

The price data set does not include any district field as the wholesale price of a commodity on a particular day is recorded with the commodity regardless of the district from which it comes. This required that the price and climate data sets be joined at a row level, to not lose the

granularity of the stations in the climate data set, but to determine how to collapse the data to a single value related to each station in the price data set. The method used was to combine the climate data by day, using the average of the temperature, precipitation, humidity and wind speed at each reporting station on each calendar day, which resulted in one climate record per calendar day. This is a defensible reduction in the complexity of a first join, but it is also a real analytics cost in that it assumes that weather conditions in Nepal are the same on any given day, across all districts in the country, which is also not true – that a disruption in supply due to the weather in one supplying district is the same as the weather across the whole country. This limitation is inherited by all Dashboard views built on this combined data set as it is deconstructed and discussed.

The first merge done on the Date field with the price table as the base table resulted in 136,202 missing values in each of the four climate columns. This was not a mistake made by a join, instead the date ranges of both tables were printed and was compared directly (price data is up to September 2023 and climate data is up to December 2019). In this instance, the lack of values was covering issue, not a match issue. At this stage, one could either must fill in values of climate variables for the missing period 2020–23 or only retain the common period of those two sources. Four years of imputation were done to preserve four years of price data, but this was considered riskier from the analytical perspective because any price-climate relation based on an imputed climate would be partly synthetic. The price data was thus only available for the price to 31 December 2019 and the merge was repeated. Checking whether there were any nulls in any column of the resulting table was again successful and a check of the number of rows in the two tables, one is the price table after it is truncated and the other is the resulting table after performing the left join, matched and revealed that no price table rows were missing from the left join operation itself.

Note that the interactive views created on the combined dataset are only able to comment on the 2013-2019 range, even though there are 4 more years of documented price volatility in the source price data. This limitation is the same as noted in the background research portion of this report and is now known to be a specific documented decision to process the data.

5.4 Feature Engineering

In Python some calculated fields were made before exporting to Tableau to make sure that there wasn't duplicated logic and that these fields were available on every worksheet. The Date field was used to extract the calendar attributes (Year, Month, Month_Name, Quarter). A Season

field was calculated based on a mapping fixed with calendar month (**December to February = Winter, March to May = Spring, June to August = Monsoon, September to November = Autumn**). It is obvious that the onset and end dates of the monsoon are not the same in Nepal and the use of a monsoon season label on the calendar month will lead to misclassification of the early or late monsoon years. It was maintained in the same way throughout the entire multi-year data set and applied in a consistent and transparent fashion. The Price_Level field was created by segmenting the average price feature into **5 segments (Very Low, Low, Medium, High, Very High)** that correspond to the colour level used in the treemap of the dashboard.

A new field was then created by a function called Commodity_Category, which was based on the Commodity field and used a keyword matching function, resulting in a new field that contained the eight named categories of commodities, plus an "Other" catch-all. The categorization wrote here is different from the 16 categorization (Root & Tuber, Leafy Greens & Herbs, Grain, Gourds & Squashes, Beans & Legumes, Animal Protein & related categorization) that is actually referenced in the live Tableau workbook, and the categorization on the dashboard is different beyond what was achieved at this preprocessing stage. Two other fields are not created in this pre-processing; they will appear in the dashboard: Price_Spread_Pct, Likely_Import_Origin. They are not exactly described in the material available for this report but are mentioned here because of the lack of traceability in the final dataset, rather than to be written as an explanation on top of the existing report: there is no evidence of such an explanation.

5.5 Validation

In addition, the one date (16 June 2013) in several different commodities was extracted and it was confirmed that the temp_2m value was the same in each commodity, thus ensuring the same daily climate value (temp_2m) was broadcast on the same date in all commodities, not different or duplicated. It's a trivial check, but one that makes the difference between a merge which has been verified to be correct, and one which has been assumed correct, and this is reported for that reason.

6. Dashboard Design Justification

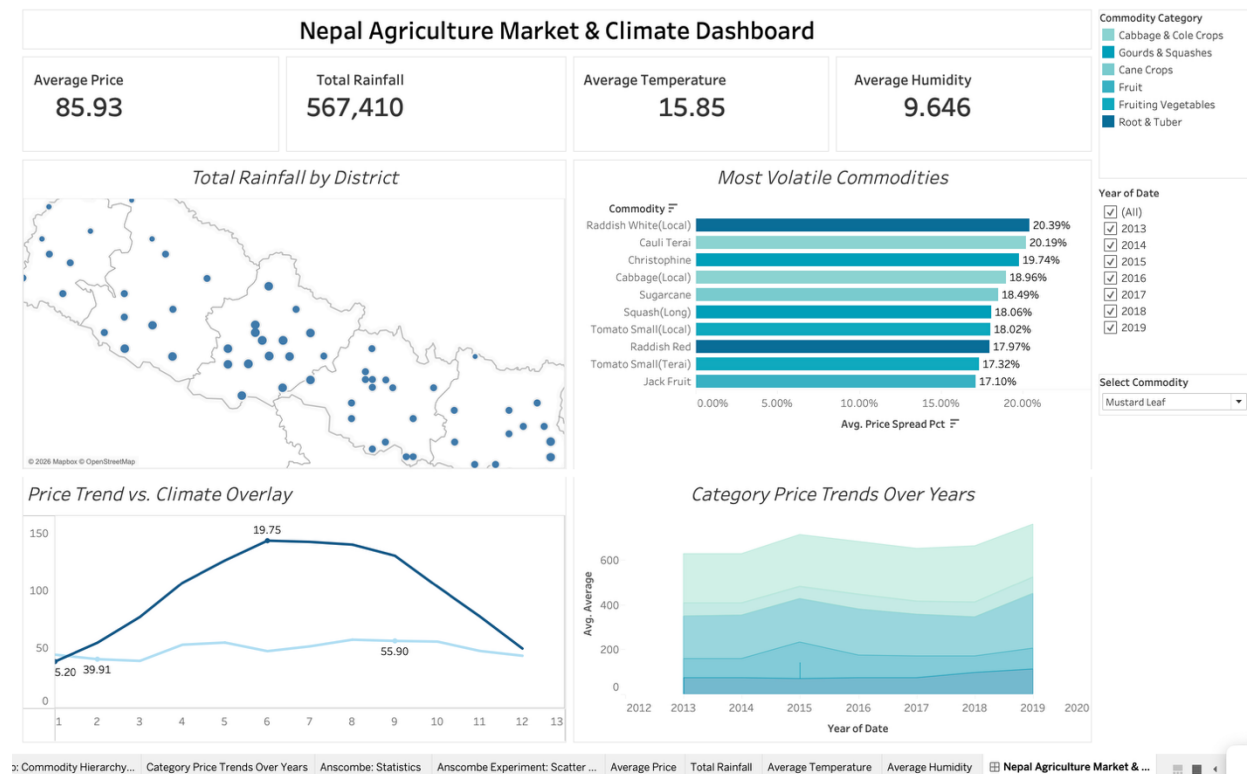


Fig: Nepal Agriculture Market & Climate Dashboard

Each of the visualizations in the final dashboards is explained and then a special section on the Anscombe demonstration, which was added but not included in the interactive dashboard canvas. The following subsections explain the purpose of the visualization, the decision that it assists with, why it was chosen rather than some other alternative that could have been used, and how it was *evaluated against Cleveland and McGill (1984) accuracy in graphical perception, Tufte (2001) data-ink ratio, Gestalt grouping, accessibility and decision support*. If a decision in design does not align with the principles that would be suggested by these principles, the lack of alignment is clearly stated, just as it would be in a report with no lack of alignment.

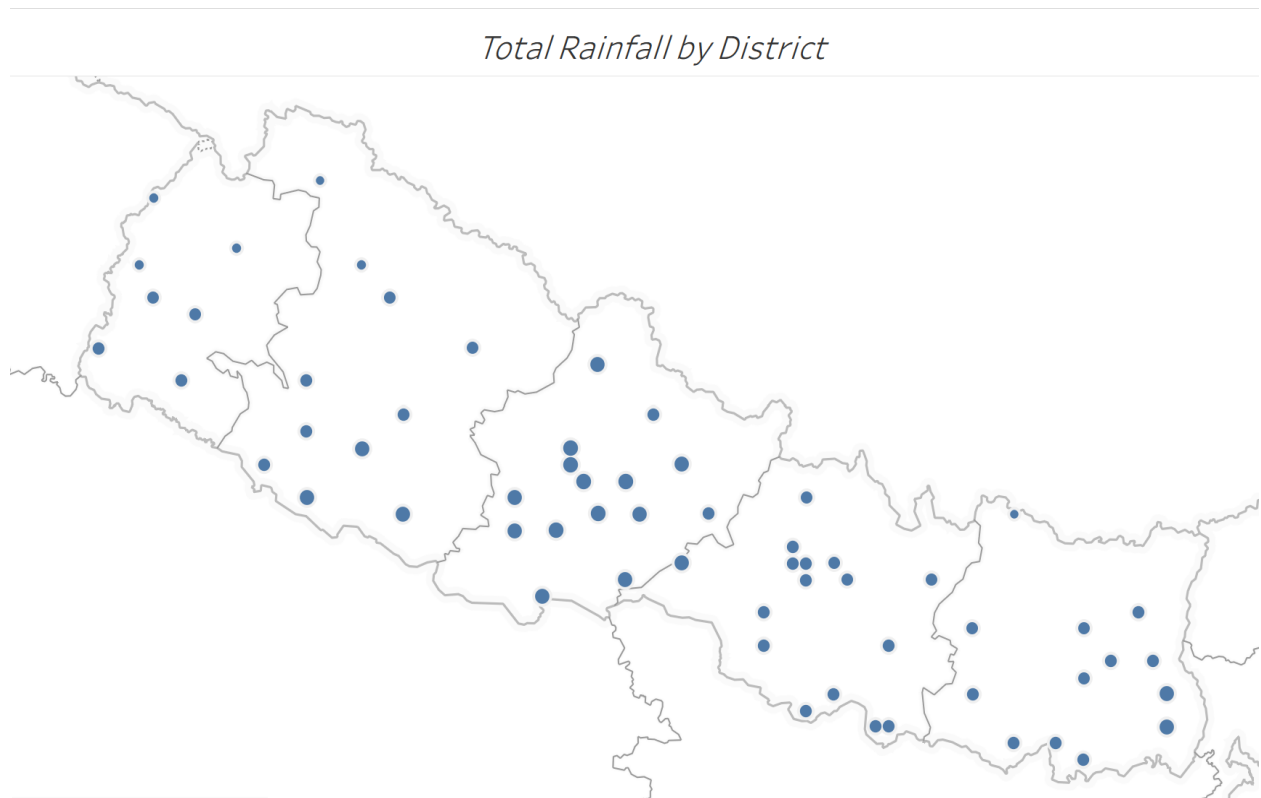
6.1 KPI Summary Row (Average Price, Total Rainfall, Average Temperature, Average Humidity)

The data is helpful to interpret using the following KPI Summary Row (Average Price, Total rainfall, Average temperature, Average humidity). Audience/Purpose choice. These four text-based indicators provide information to the viewer on the scale of the underlying data set: averages of price (85.93), temperature (15.85), humidity (9.65) and taken directly from the merged data set described in Section 10. They are not a decision that is made alone but serve as a reference for all subsequent chart decisions. Alternatives to chart selection and rejection were rejected. One aggregated number is the proper way to represent one aggregated fact, but one bar or gauge is just another piece of chart junk, according to Tufte (2001). Cleveland & McGill (1984) is not directly involved here because at this point, there is no request from the viewer to compare marks to each other.

Gestalt and layout. Two of the 4 indicators are deliberately applied in a common "light fill" and all in a single, framed container, in line with *Gestalt principles of proximity and enclosure*, otherwise the 4 numbers might be competing for attention with the charts below it and would not be read as a single unit of meaning.

Accessibility and limitations. The basic contrast requirements of the KPI text are met with: Black text on a light neutral fill. None of the four KPIs has a trend indicator or a comparison baseline (such as "versus the previous year") and it is therefore not possible to determine from the KPI row alone if there is a rise, fall or level. This is not a design choice, but a restriction, and will be examined later.

6.2 Rainfall by District (Map)

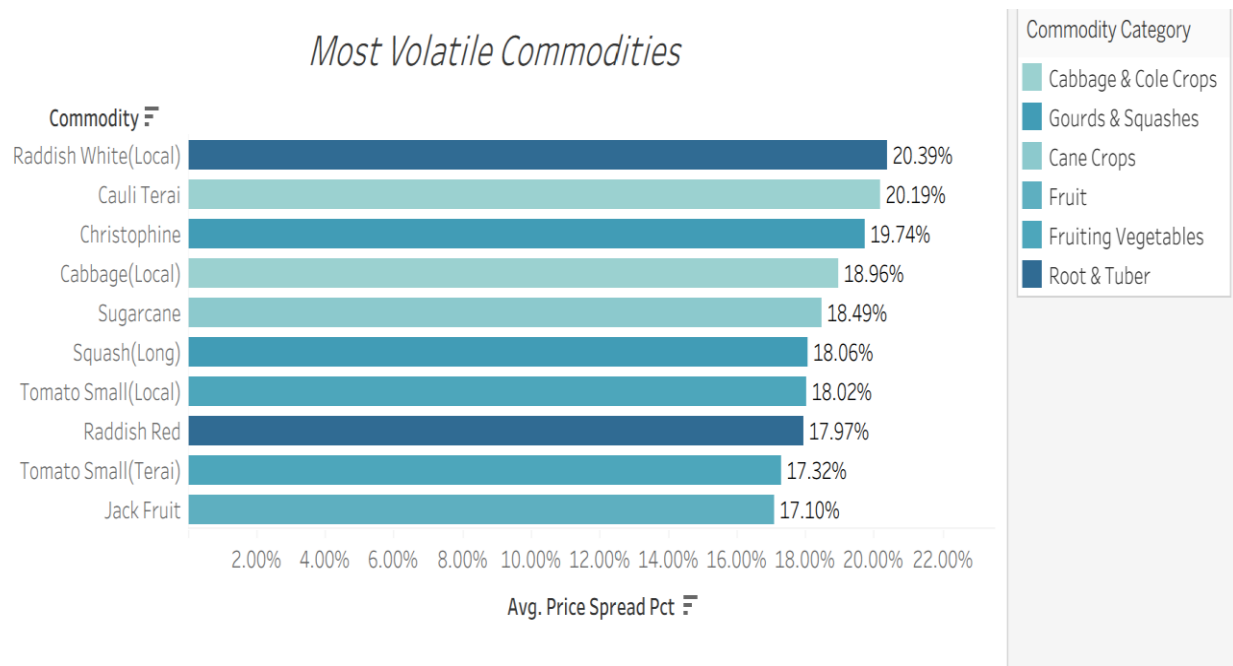


Audience and purpose decision. This view recognizes KFVMDB's sourcing-risk decision up-front because if the identification of the wettest districts in which supplies would be at risk, alternatives to sourcing arrangements may be considered before the risk of disruption. Alternatives to chart selected but not used. Alternatives to chart not used. The under question of 'which districts' was not chosen because it is most easily answered by a ranked bar chart or table, but rather because this is a spatial question that is best answered by a map via position, rather than a ranked bar chart or table, which *is the most accurately judged visual channel (Cleveland and McGill 1984)*. The total precipitation is presented in terms of point size, which is one of the most basic elements of quantitative judgement, and is more suitable for the purpose of the map in this case than comparing two similar numbers.

Colour, access and restrictions. They are all rendered in a single colour, just the size is changing, eliminating the risk of being unable to use the colour ramp for a quantitative judgement the size already provides. The cross-view coherence of the dashboard, however, is less robust than in the data-ink data, with no unnecessary basemap detail, gridlines, or decorative styling (only what is required in this map to locate a district in Nepal); no filtering

of the map by commodity, or by the dashboard's Year control, hence the map displays the full study period when a viewer specifies a different year elsewhere on the dashboard and is considered a limitation.

6.3 Most Volatile Commodities (Bar Chart)



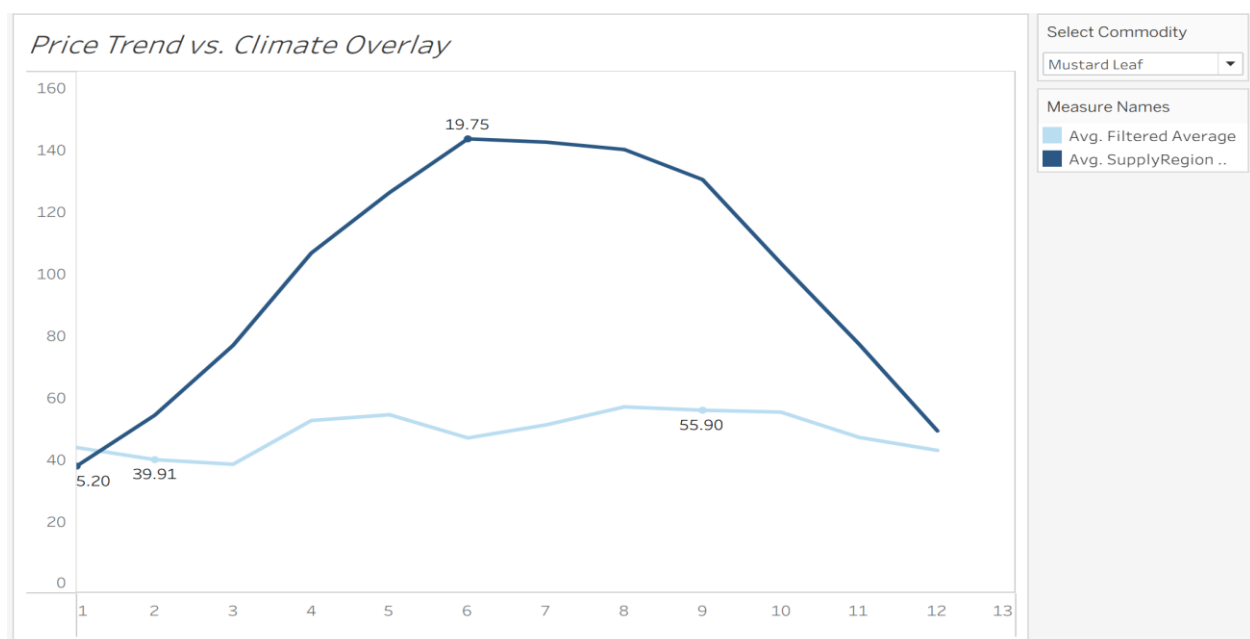
Decision on purpose and audience. A ranked bar chart that answers the question: which commodities should decision-makers monitor is used in this version, rather than a ranked list of commodities by price level, as was used in an earlier draft. Price level is a lot less informative to this audience: a commodity can be high priced and not volatile, or it can be low priced and volatile but tracking its volatility is what matters to the weather exposed. The fact that commodities such as White Radish (Local) and Cauli Terai are best looked at as an item to be paid the closest attention in the present ranking would not have been revealed by a price-level ranking.

Alternatives to chart choice were rejected. The encoding of position along an aligned axis is the most appropriate when the task that the bar chart represents is a ranked-comparison task such as the case with a horizontal bar chart, according to *the principles of Cleveland and McGill (1984)*. A task in which the majority is a rank comparison task would need a treemap encoding, which would require an area-based judgement.

Colour and Gestalt. The list is then *arranged and colours are clustered by similarity* to the viewer's sight so that the viewer can see, for instance, that all the root vegetables are grouped on the volatile end of the list even though he or she doesn't have to read the axis labels individually: *a direct application of the Gestalt principle of similarity*. This is also the obvious drawback of this chart: The colour ramp used is sequential (blue to teal) and was originally designed to be used in an ordinal field on this chart but is used across 16 unordered categories. The ramp is not well differentiated visually in several neighbouring categories, and so does the submitted workbook not have an alternative that is qualitative and colour blind safe.

Interaction. This worksheet adds a highlight action that will copy a selected category for a selected commodity to other marks in the same view but not (in this version) to the line or area chart the selected commodity is displayed with.

6.4 Dual-Axis Line Chart: Price Trend vs. Climate Overlay



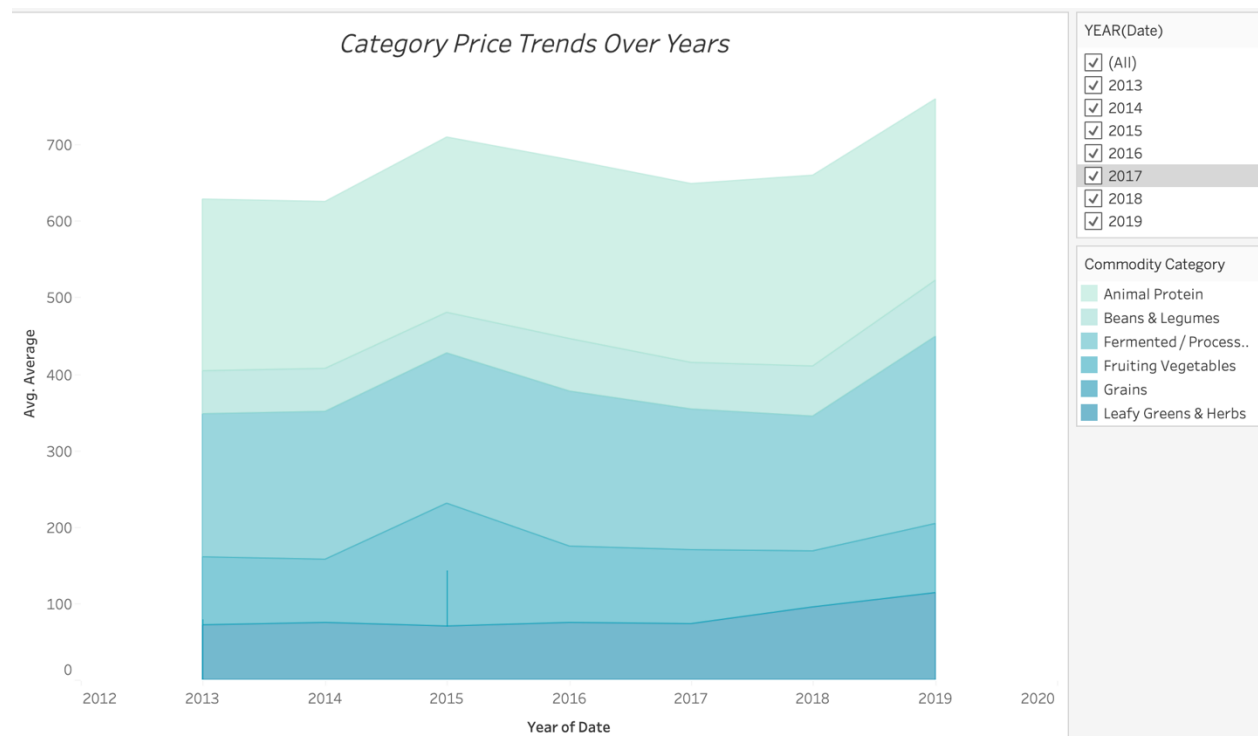
This is a line graph that displays two sets of data on one graph, one vertical line showing the price and the other showing the climate. Determine purpose and audience. This is the perspective on timing and says that yes, the seasonality of price is close enough to the seasonality of minimum temperature to justify providing a price advisory before the price changes than afterwards.

Chart choice and alternatives rejected. The reason for the choice of dual axis line chart was that the axis representing months was the same ordinal (integer) axis and the two measures plotted were continuous (real-valued). For this reason, two axes were selected as price and temperature were measured in incompatible units: A normalization step would introduce the danger that the viewer would not have a clear impression of the relative magnitude of the changes in each series, **directly related to Tufte's (2001)** warning against unnecessary transformations that distort the data.

Colour and accessibility. The two lines are not rendered in a single colour at two levels of saturation, but in two distinct colours; this is a deliberate decision as two shades of the same colour would not have clearly distinguished the price series from the climate series for a colour-vision deficient viewer and maybe for anyone at first glance.

Limitations. There are two data labels in the way, one near the bottom-left of the plot area (2.20), and another one (35.55) that just happens to overlap at the point where the two series are closest together making it less, not more readable. *This is one of what Tufte calls "chart junk"* that should be avoided, and a simple targeted fix (limit the number of marks labels are attached to the series maxima or just remove them altogether and have reading on a tool tip) has been identified but not yet implemented.

6.5 Category Price Trends Over Years (Stacked Area Chart)



Purpose and audience decision. This view presents a more long-term planning question than the KPI row or the dual-axis chart: what has been the price contribution evolution within the planning period for each commodity category, which is more relevant to the Department of Agriculture's policy-planning horizon than to the decision on sourcing during a specific season.

Chart choice and alternatives rejected. A stacked area encoding was selected to display the trajectory of each category of prices as well as an approximate idea of each category's contribution to the overall average price, in a stack. This is a true trade-off, not simply an obvious best choice: stacked areas allow easy and accurate reading of the trend of the top band; in comparison, every band below it is more difficult to read precisely, because they all are not drawn on a common baseline, which is a disadvantage of stacked areas over the alternative small multiples form that **Munzner (2014) would recommend for precise comparison of multi-series**. This was deemed to be acceptable as the purpose of this chart is to communicate overall directional pattern which would be more effectively communicated on this dashboard by the ranked bar chart.

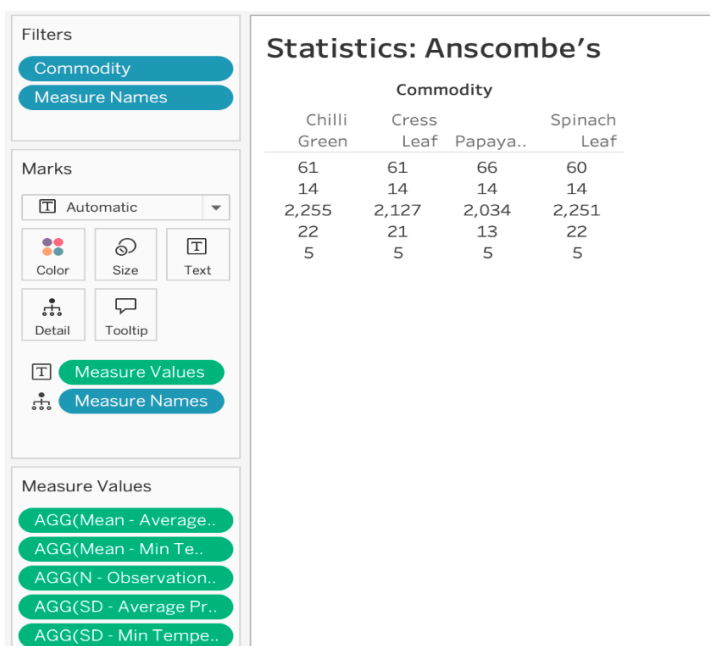
Interaction. This worksheet is the original worksheet for a Boolean "Show Insight" parameter that was used in a development parameter action and was originally part of a "progressive disclosure" design that was supposed to show supporting analysis when it was selected. That

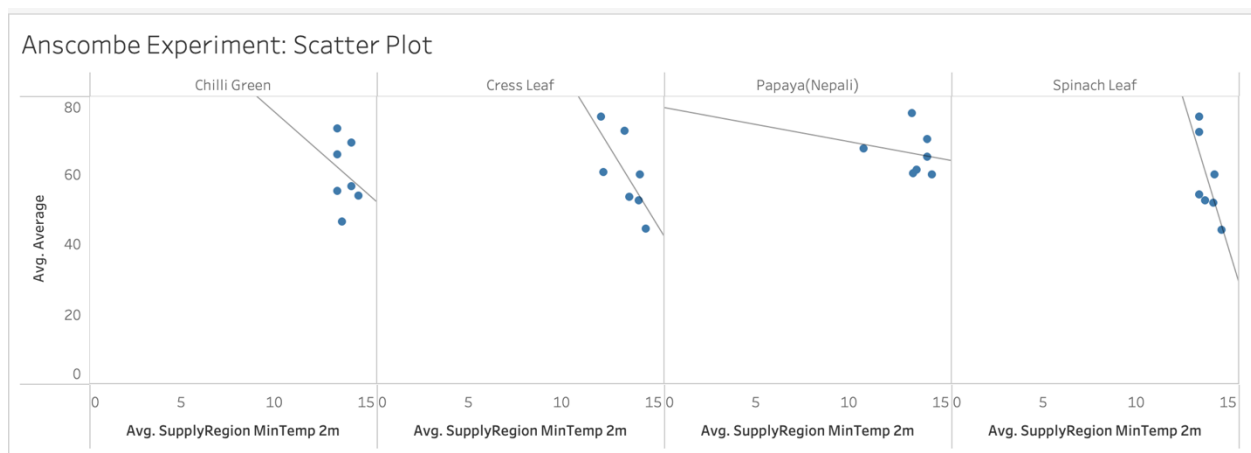
parameter action is still defined in the workbook, but there's no Dashboard component bound to the action of the parameter, so it doesn't seem to affect the Dashboard in the delivered build. This is reported here as a design experiment that was not successfully completed but is honestly reported as such technical learning.

6.6 Interaction Design Across the Dashboard

Four actions for the dashboard are defined in a workbook. *There are two default filters, one that is applied to all fields, applied to all sheets when selected - applied to all sheets (a filter action and a filter highlight action).* They provide the dashboard with a basic level of cross-filters without having to first understand which cross-filters to which each of the dashboard's charts correspond. When a third action is identified, it will be scoped to the Most Volatile Commodities worksheet when it is selected. Note: that these *two controls (Year filter and Commodity parameter) can be found on the dashboard canvas but will only impact on the Category Price Trends Over Years worksheet* and not all the other worksheets that are compatible with the date field. Therefore, if a viewer changes this on the dashboard, they will not notice the change in the map, bar chart or line chart. This is not an interaction design feature, but as part of the entire dashboard level filtering as evidenced by workbook evidence.

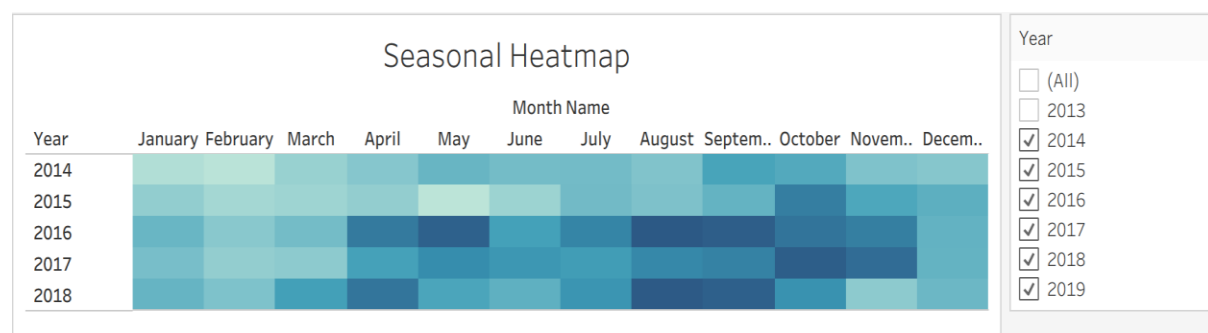
6.7 Anscombe's Lesson: Supporting Analytical Worksheets





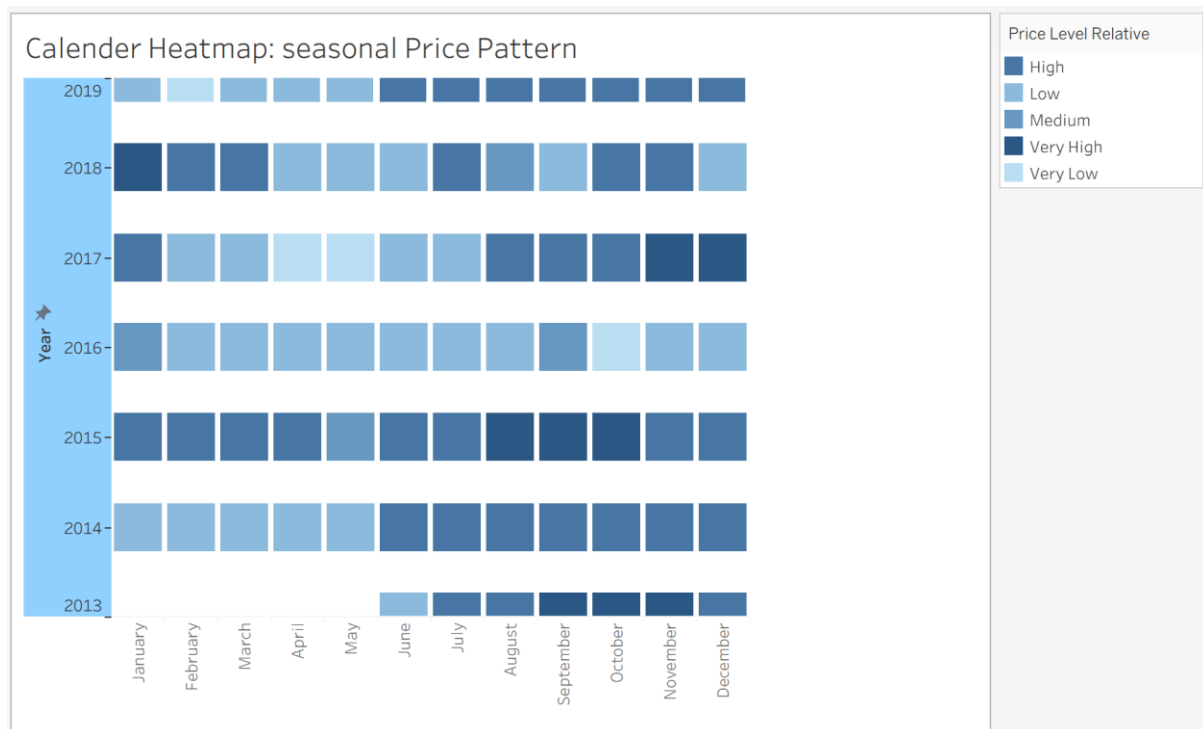
Two worksheets, Anscombe Experiment: Scatter Plot and Anscombe: Statistics were written to highlight the main point of Anscombe's (1973) paper: that two sets of summary statistics (e.g. means and standard deviations) can be very similar but represent very different relationships and that a table of summary statistics is not sufficient to determine whether a relationship is linear, nonlinear or outliers dominated. The statistics sheet calculates the average price and the standard deviation of the price, the number of observation of the average price, the average temperature and the standard deviation of the temperature and can be filtered by the commodity and by the measure; the scatter sheet displays the plot of the average price and the minimum temperature and is capable of being filtered by the commodity and by the measure and faceted by year. Reading together, the two worksheets enable a viewer to see not only the summary statistics of a commodity but also the commodity's scatter shape side by side and to clearly observe that two commodities having similar summary statistics of price-temperature correlation can have visibly different scatter shapes exactly the sort of phenomenon the underlying calculated fields were originally designed to show.

The interactive dashboard canvas was intentionally left without two worksheets. It's not an error – the key audience of the dashboard, the KFVMDB staff, who are making fast, action-critical decisions, should be supported by limited decision-oriented views, rather than two additional exploratory views that are more likely to be demonstrative and analytical, and less likely to be used for action. This compromise is a reality though. The assignment criteria for this component specifically require that the Anscombe-like demonstration be a prominent and



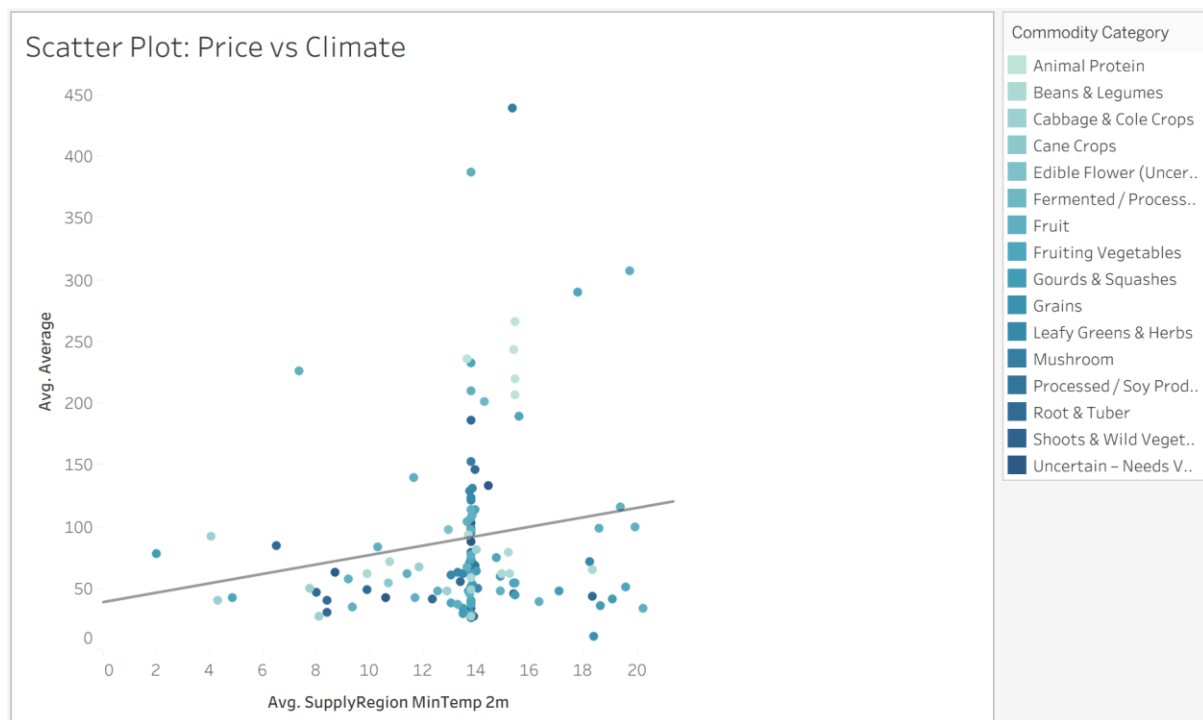
Average price coded in each cell (sequential colour ramp, colour blind friendly). **Chart choice.** When the question is about "when does price peak", and the data is a continuous quantitative variable, a single-hue sequential ramp is an appropriate encoding as the darkest cell is the highest value, and the lightest cell is the lowest value without having to compare actual numbers, and this type of question is perfect for a heatmap.

6.10 Heatmap: Seasonal Price Pattern



Same year-month grid but coloured according to a continuous measure of commodity relative prices, but with a five-level, ordered palette of colours (Very Low through Very High) and a colourblind friendly palette, patterned in the order of the commodity relative price level. **Chart choice.** This field is not purely categorical, so it was important to design the colour ramp to be in logical order or there would be five different colours without any obvious "lower than / higher than" relationship to the viewer which would defeat the purpose of an at-a-glance reading of the map. This is done in conjunction with the average price heatmap, so that two complementary readings of seasonality are available-one of magnitude, and one of relative tier which when combined tells more of the story than either one does individually.

6.11 Scatter Plot: Price vs Climate



Average Price by Min supply-region temperature, with commodity category data coded as colour bands (a colour-blind friendly palette was used), showing likely origin for imports.

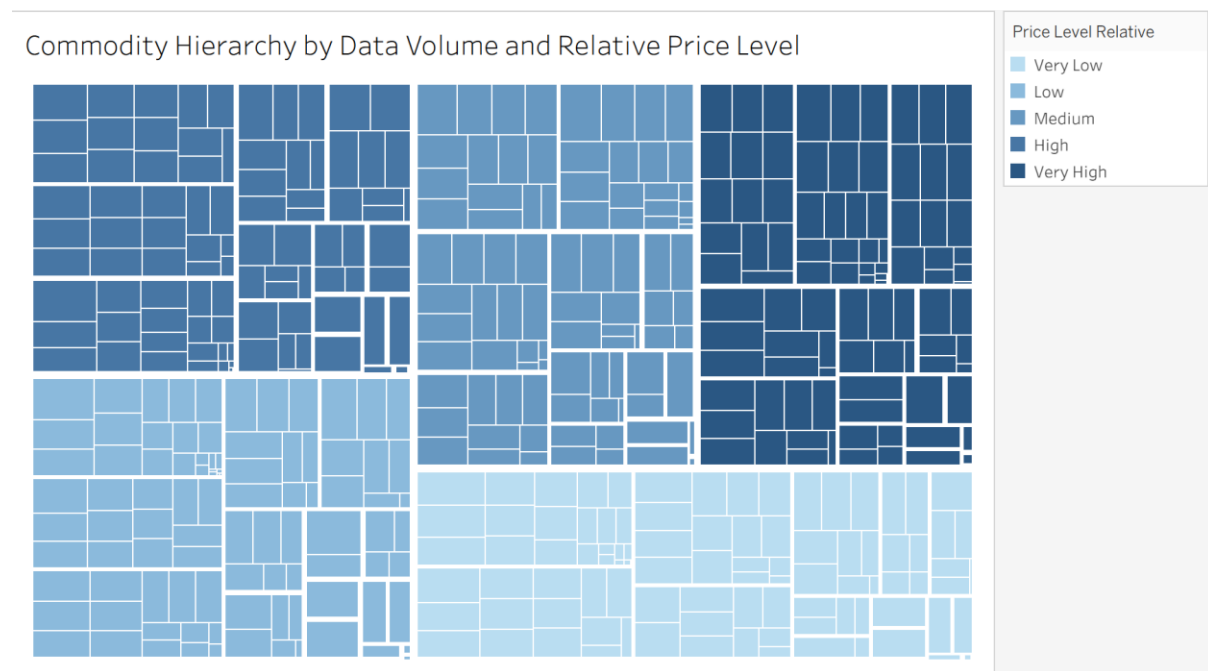
Chart choice. The question in this case is about the form and strength of the relationship between two continuous variables, which a scatter plot can better handle than a bar or line chart. Here, an adjacent text label is optional like it is in other scatter plots and therefore having a chart with an easily distinguishable colourblind-friendly palette was more important than in any other scatter plot in the workbook, if the viewer can only tell which category a given point belongs to by looking at the colour.

6.12 Slope Chart: Seasonal Price Shift (Monsoon–Winter)



Average price is displayed across 2 seasons and colour by commodity category, in a colour-blind friendly palette. **Chart choice.** The goal of this view is to provide a single visual sense to a viewer of the direction and magnitude of the price's change between two specific points within the same year, without having to "do" two lookups and then subtract one from the other. Here, the colour encoding was important as many categories could be going in many different directions at the same time, so the viewer must be able to follow an individual line crossing others.

6.13 Treemap: Commodity Hierarchy by Category



Commodities nested within their category, sized according to the number of different commodities in the category, coloured according to the relative price tier with a colourblind-friendly ordered colour palette consistent with the price tier's natural sequence from Very Low to Very High. **Chart choice.** A tree map was selected because the question has two components: how much of the data set does this category represent, and what position does it occupy on the price scale, and a tree map is one of the few types of charts that will encode the part-to-whole relationship and the value comparison together into a single view: with area representing the part-to-whole and colour representing the value comparison.

7. Key Insights & Recommendations

7.1 Insight: Price volatility is unevenly distributed across commodities

Evidence: The Most Volatile Commodities view is a list of 128 commodities which are sorted by their average price-spread percentage. The price-spreads are concentrated around the top ten, which range over a relatively small period (20.19% to 20.39%, with White Raddish (Local) at the top of the list and Cauli Terai at the bottom) and the price-spreads overall are much wider in the underlying dataset that was used in the dashboard.

Interpretation: A small and recognizable subset of commodities is responsible for proportionately more than a small share of the volatility in a national vegetable-inflation figure as a single number. The two commodities mentioned above are further proof that the changes in volatility can be big between varieties of the same commodity, when a national statistic would only account for one commodity.

Business implication A KFVMDB advisory, which is described in the literature reviewed, will systematically under-warn on items likely to see vegetable inflation, and over-warn on those items that, in relative terms, will see relatively low-price changes.

Recommendation: The Board should base sourcing conversations with supplying districts on the ranked top ten volatility list, as opposed to the aggregate price index, and consider reviewing and updating the list annually using the Commodity parameter in the Dashboard and on the individual trend of each commodity flagged on the top ten volatility list before communicating an advisory.

7.2 Insight: Price and minimum temperature share a broadly common seasonal shape with a lag

Evidence: The overlapping of two axes shows that both the average price and the lowest temperature rise from a low around month 1 or 2, to a maximum around month 6, and then decline towards the end of the year, but not necessarily during the same month.

Interpretation. It is consistent with a seasonal transmission mechanism where warming conditions and the onset of the monsoon pressure slowly ramp up during the season, but is also consistent with multiple interfering seasonal factors, including the planting and harvest seasons, which co-vary with, but do not necessarily match with, temperature. This association is not cause and effect and there is no calculation on the dashboard that would differentiate between the two.

Business implication. To suggest that such a price change is based on temperature from this graph would be taking a huge risk and result in an incorrect cause/effect scenario.

Recommendation. The dual-axis view should be used by the Board as a reminder to carry out further seasonal monitoring but not as a justification for an advisory and should be accompanied by the qualitative supply-chain reasoning that is currently used by the Board's

information officers (described in the news-source review in Section 7) when communicating on this chart, rather than being treated as a statistically established causal claim.

7.3 Insight: District-level rainfall is spatially concentrated

Evidence. The Rainfall by district map shows a more concentrated area of large points (high rainfall amount) in the eastern and central districts compared to the western districts.

Interpretation. However, for any commodity mostly supplied from eastern dense areas of Kalimati, the eastern dense areas of Kalimati are more sensitive to the rainfall risk compared to the western less dense areas in Kalimati.

Business implication. This district concentration would be totally hidden if one average rainfall value is used for the country as a whole or for the entire valley.

Recommendation. The map should not only show a high rainfall area but a district in that high-rainfall area that supplies a particular commodity, which is why it is important to cross reference with the Board's current documents of supplying districts for each commodity (as identified in the original project proposal risk register as a data preparation step).

7.4 Insight: Category-level price contribution has grown unevenly across the study period

Evidence. The Category Price Trends Over Years area chart displays the top band in the area chart in a widening band, increasing from about 400 to 600 over the visible years, and smaller bands in the area chart increase over the years.

Interpretation. The pressure on prices is not the same for every commodity group; more of the total average price change over the study period (a policy timescale and not a single season) will be at the top of the chart.

Business implication. If the policy maker only looks at the multi-year price trend, he or she would not be able to determine which category is responsible for the growth in that trend.

Recommendation. It is important to note that the category which contributes most to the total indicated in this chart is not a one-off recommendation from this report but should be a regular

agenda item at the department policy review meetings, and reviewed annually when underlying data in the dashboard is updated.

7.5 Insight: The 2013–2019 dataset window cannot speak to the most severe recent events

Evidence. The end of the climate data set for the dashboard is detailed in Section 10, due to coverage. This combined dataset in the dashboard is limited up to December 2019, because of the coverage limit of the climate data set.

Interpretation. The price-climate relationship shown on the dashboard cannot show from the data on it the price-climate relationship that is expected to be shown in more extreme years when checking for continuity (2024-2025).

Business implication. No communication from any KFVMDDB claiming to represent a 2024-2025 event with this dashboard as evidence would be accurate in stating the temporal coverage of this dataset.

Recommendation. The other recommendations in this section rely on the underlying data set actually extending into the future with which a decision-maker is having a decision made about, so that adding the year 2020 onward to the underlying climate data set should be the highest priority data investment for any continuation of this project, before considering other visual features, in particular.

8. Ethical Considerations

8.1 Data Privacy and Aggregation

The price data do not contain personally identifiable information; not in so much as it identifies a particular trader, but rather a record of wholesale price data concerning commodities; the climate data do not identify data or readings from a particular station or reading, but rather a record of data or readings from the station. In addition, the station-level climate readings are aggregated to a single daily mean at the national level in the aggregation step of the dashboard thereby further reducing the risk of re-identification at the expense, already mentioned, of discarding the station-level precision. The privacy risk in this instance is low, but it is worthwhile to explicitly state that the privacy risk and the aggregation-bias risk are two

different properties, and that the choices of aggregations in the dashboard were not based on the privacy risk but were instead based on a constraint in data integration.

8.2 Aggregation Bias and the Risk of Misleading Visualization

One possible source of aggregation bias with direct ethical implications is the calendar-month boundaries of the Season field which might trigger KFVMDB advisories to the wrong time of year if the actual monsoon is early or late. Similarly, the stacked area chart can only be interpreted accurately for the highest band, and someone looking at it purely on a graphical basis may assume that the lower band's risks are reduced as well. They are both listed here because they are part of a specific design decision and not as blanket disclaimers; the ethical part of this assignment is to point to each concern in this dashboard rather than to the visualization practice in general.

8.3 Correlation Versus Causation

An obvious location of causal risk, based on the dashboard, is what insight can be gained about the co-movement of price and minimum temperature. Both axes are seasonal, so the seasonality of the data in the two is very noticeable and a person who tends to see a causal story in the relationship will see one; they are not even subjected to any statistical test of the relationship, and other features of the agricultural cycle mentioned, further muddy the water. They don't appear anywhere on the dashboard canvas, but in this particular risk, they have been placed onto the Anscombe worksheets, as it is easier to add a warning that a summary relationship's statistics do not reliably indicate the true shape of that relationship if it appears on the canvas. Truly, it's also an opportunity lost to show this warning to the readers who are mainly there to verify the accuracy of the price-climate overlay.

8.4 Colour Accessibility

The Colour palette chosen for the bar chart is sequential, blue to teal, and is not a good choice for discriminating between the sixteen unordered categories it is used for, nor has it been changed to a colour-blind safe qualitative palette in the delivered workbook. The accessibility failure is directly identifiable and has an identifiable population to which it impacts: A viewer with red-green or blue-yellow deficiency will have difficulty distinguishing the next step in the ramp from the other steps when compared with a viewer who does not have the red-green or blue-yellow deficiency (Wong, 2011). This is expressed as a problem not a solution, as per evidence within the workbook text.

8.5 Representation of Marginalized Farming Communities and Dataset Limitations

The price dataset is limited to a single wholesale market, which does not capture data from many smallholder farmers in often remote and under-developed districts who supply the market; nor does the price dataset necessarily benefit from the price signals visualized on the dashboard. The map of Rainfall by district may not include the population that the supply chain depends upon if used by a user who assumes that the map represents the affected population in the affected district. This is not specifically stated in the claims of this dashboard, but it is a fact of what data that is available in the public domain can represent and not an inherent limitation of the dashboard itself, it is worth highlighting that a decision-maker who uses this dashboard alone may be misled to believe that the coverage of data is the same as the full picture.

9. Lifelong Learning Reflection (TGC4)

Role	Member	Actual Work
Team Leader	Sapana Tamang	Built the dashboard, KPI summary & the chart layout.
Research Domain Lead	Anish Bhandari	Sourced the datasets, Took lead of literature review and the research & visualization gap. Helped with the reporting.
Data Engineer	Osin Dhamala	Pulled different strands: merging data, dashboard & the report. Included the DV principles in all of them.
Visualization Designer	Osin Dhamala	Built the varieties of the charts as per the project's requirements & worked on Tableau.
Documentation & Presentation	Karin Maharjan	Helped with the report structure, table of contents and the self-reflection contributions.

9.1 Self-Directed Learning

The final dashboard included some functions that were not included in the module's curriculum and were explored outside of teaching. One such example is the dashboard action model in Tableau. The difference between the "all fields" action scope and the "field" action scope and the distinction between an "on select" action, and an "on clear" action, wasn't intuitively apparent from Tableau's default pop-up, and required iterations of testing different actions in order to make the determination that the time to read the documentation was not worth the effort.

One more ambitious attempt at self-directed learning was to try to develop an interaction that would be "progressive-disclosure"-style: A Boolean parameter to display an analytical panel on request, but never permanently take up screen space in the dashboard. This included understanding that a parameter action is not a filter or highlight action, but rather an update to a stored value, and how to add a binding on visibility conditions to the container not to the

worksheet that triggers it. This second binding step is not completed in the workbook delivered, and the parameter action is still defined, but remains without effect. This does not report the result of the creation, but as a learning outcome of the attempt: to learn why to use two-step binding and where the implementation went wrong; this is because the team is learning what they need to know, whether the feature is created.

A second use of self-directed learning was comparison of source date ranges – another debugging skill that the team had to learn rather than be taught.

9.2 Self-Inquiry in Learning

While building the dashboard, the team asked itself again, as it was creating these charts: Does this chart help a KFVMDDB or Department of Agriculture stakeholder make decision or is it merely displaying data that is available at the time? It was this question that led to the switch from Top Commodities by average price to More Volatile Commodities ranking: the question 'which commodities are expensive' is descriptively interesting but might not be of interest to the decision makers and the new question 'which commodities should be monitored for weather exposure' is the one the seasoned audience is asking.

Continually reengaging with the scope of the dashboard, as reported in background research section of this project, was a second continuous source of inquiry during the project whether the scope of the dashboard was broad enough to be 'useful but not confusing', or whether it was becoming 'confusing enough to be useless'– interaction with Mr. Anmol Adhikari (the module leader) was ongoing throughout the project, constantly testing the scope of the dashboard and encouraging retention to Gestalt grouping, Tufte's data-ink principle and Anscombe's lesson throughout the design process. This sort of investigation led to the decision to make the Anscombe demonstration a separate analytical worksheet rather than it being the focus of the dashboard, at the cost of not focusing on it?

9.3 Self-Assessment for Learning

The team decided to use the marking criteria provided in the module as their own as well as use this as a working checklist when developing the module. The team's self-rating of how well the dashboard meets the interactivity descriptor in the rubric is low, not yet in the Outstanding band, before they have used external feedback, external feedback in this instance being by comparing the dashboard to the Beginning to Outstanding band descriptors in the rubric.

The self-assessment directly contributed to the addition of the 4 dashboard actions and to uncovering the lack of filter connection in the Year action filter and no filter connection in the Rainfall by district map during the second round of self-assessment, following the completion of the interactivity work. This iterative process (that is, test a rubric descriptor, identify the gap, fill in the gap, retest) has resulted in a workable dashboard that is clearly headed in the right direction for meeting more of the requirements of the rubric than the original, but also leaves an honest, well-documented set of “gaps” that do not leave a false sense of completion that the workbook evidence does not support.

9.4 Self-Reflection

Each team member's reflection should be added directly beneath their name below, in their own words, addressing what they personally learnt during the project, what they would do differently, and how the project has changed their approach to future data-work. These reflections should not be written on a team member's behalf; the placeholders below are structural only.

Team Member 1 — [Sapana Tamang]

Creating the interactive dashboard was one of the most beneficial learning experiences in this project. I found this to be a more difficult project than my previous Tableau dashboard project, as I had to work with a larger data set, needed to follow the requirements of the project and needed to create the dashboard to effectively communicate meaningful insights. It did give me the ideas to think outside the box when charting and was to some extent, able to express information to other users.

Over the course of the project, I discovered that creating a good-looking dashboard is as much a matter of knowing how to use each visualization to send a message, as it is making it look good. The ideas introduced in the course (including Gestalt Laws, Tufte's Data-Ink Ratio and Anscombe's Lesson) provide insight into the logic of visualization design and how to improve clarity and understanding of visualization for users. The lecture and lab sessions were also a means to learning. If I had any questions and if I didn't understand something our lecturer would go into much more detail and explain to me how it could be applied in practice.

At times, I wasn't sure if I designed my dashboard to the best of my ability, formatted my KPI cards and even enhanced the entire design process. Brainstorming with teammate “Osin Dhamala” helped me to perceive the dashboard in a different light to make decisions during the design process. Throughout we continued to discuss ideas, questions and encouragement in the group chat when we had a challenge.

This project helped me enhance my skills in Tableau, critical thinking and collaboration, and got me better at designing dashboards. Most importantly, it was a lesson in question and answer, feedback and continual efforts to enhance one's work as part of the learning process; and that I can continue to enhance my work in my future academic studies and career.

Team Member 2 — [Karin Maharjan]

During this project I knew that I wanted some data visualization but I never realized it was more than just creating a graph and dashboard, it was about communicating data clearly, accurately, in a way that is meaningful to the viewer and allows them to understand the information presented and act on it.

In this evaluation and final sections, I got to learn more about the techniques of data visualization, creating data dashboards, using the right type of chart to convey trends and patterns. The project helped me to gain a deeper insight into critical thinking, strategic planning and writing reports, as well as evaluating the quality and effectiveness of data visualizations.

I would also better review dashboard during the whole development process and give feedback earlier to give better products. This experience has led me to expect much more than just creating appealing dashboards from data not only should it be aesthetically pleasing, but it also should be accurate, simple to use and something that could convey a story from data. Overall, this project has given me a sense of confidence in my skills with data visualization and has sparked ideas for future projects that I may engage in, academically and professionally, which will involve data analysis and data presentation.

Team Member 3 — [Anish Bhandari]

When I began working on this project, for me, the data visualization process was more of a design challenge, so, chart type, colouring, making the whole dashboard look nice. One thing I learned while going through making this thing; there's very little of the actual work to see in the product. The most important decisions I made were in preparation for the first chart, when

two data sets didn't have an obvious common structure, and when the merge didn't behave the way I wanted, how to deal with this as a question I wanted to investigate, not a bug that I wanted to patch and go back to sleep for. What is helpful and what I am working to remove from myself from this module, that I didn't have at the initial start, is seeing the error as, make the error go away and not as, why did this error occur?".

I also discovered the hard way that a well-completed dash can be done, and it may feel like it is done, but it's as wrong as it can be. At this project, that's happened more than once: it has been matching exactly what I saw on the dash when I first put it together, until I started digging into the details of how it was being aggregated, and discovered it wasn't. I think it was a humbling experience, late in the degree for me, because I knew how easy it is to just say, okay, this is a chart, it's nice and clean, doesn't have a red warning, doesn't have an error, doesn't have anything that says, there's something wrong with this chart. Those problems taught me that checking a design is important as it is to create a design and that creating a design is not the same thing. It's possible to make a number appear confident in a nicely formatted card, but it takes a much slower form of attention to ensure that the number is correct, and I'm not sure that I gave it enough of before.

What I found the most difficult in part 1 was being able to leave things out. After all, we made more than we needed and made a few choices on what to include on the end dashboard and what to include in the report, that is the honest one, I had a few times here. I didn't feel like planning on fewer more specific images, I felt like continuing to add images. As I completed this project I had to become more comfortable with making that decision and, in some sense, publicizing it, making it more public by writing about it.

There are no skills I am looking to bring with me from this class. But I will take with me the discipline of saying I don't need to take out something that I built that I might have not really needed for the audience that it was written for, but I need to be able to say a limitation down where it belongs, instead of hiding it, or instead of checking a number, I prefer to submit a piece with known, known shortcomings rather than a piece that appears to be complete as well as overpromising its capabilities.

Team Member 4 — [Osin Dhamala]

This module of Data Visualisation has played a pivotal role in my learning, and I have had to think of data in a new way, it has certainly changed the way I think about data professionally.

In the process of this project I learned not only to be passionate about creating information that was appealing to the eye, but also that visualisation is a serious analytical tool, that regulates for accuracy, perception, cognitive load, accessibility and decision support, *that applies the Gestalt principles, Tufte's Data-Ink Ratio, Cleveland and McGill's principles of graphical perception, and that applies Anscombe's lesson on the dangers of relying only on summary statistics*. I wrote the written argument for the project, designed and developed the views (including dashboard views) and justified all my design decisions for all views, some of which I did during designing the views, against the academic literature and the theory of visualisation in terms of the written argument.

The process of linking the wholesale price data for Kalimati to the daily data on climate was eye opening and highlighted the need of quality data and data that are climate and contextually relevant to inform and guide policy decisions. There were some very hard trade-offs to be made in reconciling nearly 294,000 price observations every day with more than 883,000 station-level climate readings, such as using only the common period 2013-2019, and assuming that the weather at each station was combined into one reading for the country. I learned that yes, it took me awhile to make several decisions – what data to aggregate, what time periods to choose, engineering features, and proving in real world – to successfully produce any meaningful results – not just a pretty chart. It also gave me the opportunity to view a cool dashboard and realize that the data prep wasn't that well done, or clear. At the same time, I developed a better combination of aspects of the implementation, what audience/stakeholder needs, theoretical principles and critical evaluation, and putting it into a coherent argument, while working on making the dashboard.

This module & the group work got me to thinking more about myself as a learner, and about evidence to inform my learning. This underpinned many of the decisions made in the project, such as not putting a chart on the primary dashboard canvas but instead having targeted views that could support the audience of the KFVMDB to inform them of deciding (not exploratory). I feel that this has helped me to understand how important integrity, visual perception, the ethics of communication and accountability is to anyone who works in a data related profession and the skills I have learned in this module will transfer into my future studies and profession in a data related field.

10.Honest Self-Critique

10.1 Strengths

The first thing that comes to mind as an advantage of the dashboard is that the fundamental chart selections are clearly reasoned and defensible based on a specific stakeholder question—whereas other selections would be a part of the default behavior of Tableau. Each of the three main charts (Most Volatile Commodities ranking, the dual-axis price-climate overlay, the district-level map) provides a reply to a different stakeholder question. It is not assumed that this is the data-cleaning pipeline described, as there is a documented explicit checking of the nulls and row counts at each point; there is some time spent, though, on trimming the data set, and it was decided to leave data out to create analytical honesty.

10.2 Technical and Interaction Limitations

There are three defects in the interaction layer, not inferred. First, the Year filter is not applied as a dashboard-wide filter but rather only to the one worksheet (Category Price Trends Over Years), so changing the Year filter does not result in the map, bar chart or line chart reacting to the changes, which reduces the dashboard's ability to provide coherent cross-view exploration. Secondly, the Rainfall by district map is not connected to any other view/highlight, nor is it linked to any other widget/control, so it is not interactive. Third, the Show Insight Boolean parameter was not connected - this is not a working feature, but rather an incomplete reported as such.

10.3 Visual and Accessibility Limitations

The colour scheme of the Most Volatile Commodities chart is a ramp of blue and teal colours – a sub-optimal (and potentially confusing) colour mapping of 16 unordered categories – not a step-by-step mapping of the 6 categories that was provided in the delivered workbook, which would be a specific accessibility problem as opposed to a general one. The data points near the minimum values along the dual axes of the line chart are a legitimate small example of Tufte's "data-ink principle": labels should be used sparingly.

10.4 Dataset Limitations

Not all events that are reported in the background research of the report are included in the dashboard, as the dashboard only focuses on events that are more moderate on the supply

disruption side of 2013-2019. The national daily aggregation for each supplying district removes the spatial variation at each station in the dataset, a compromise necessary for making a working join, but not for being able to make precise statements about the climate conditions in the district being addressed by the view. There are also limitations to the project's documentation practice: Two of the fields on the dashboard (Price_Spread_Pct and Likely_Import_Origin) have not been documented in the project documentation available for this report, and they were not traceable.

10.5 The Anscombe Scope Decision, Reconsidered

The obvious and prominent requirement is not just that the demonstration is on a takeaway, it needs to be on the canvas of the dashboard as well. Revisit here as a reflection on self, rather than a design rationale:

A dashboard with a smaller, well labelled Anscombe panel even if some of the width of the KPI row was lost, would likely have satisfied the explicit requirement of the assignment and the goal of the dashboard to support decision making better than the current version of the dashboard.

10.6 Alternative Design Choices Not Taken

A small-multiples view of the Category Price Trends view (instead of a stacked area chart) would have allowed the category price trend to be as comparable on a real level as it would be on the highest level of the stack—but would have required more dashboard space to display each category that is shown. This trade-off was not discussed since the stacked-area version was built and isn't a trade-off that has not been proven by data and evidence to be an actual alternative, but rather a trade-off that was not tested empirically in respect of users' understanding of it.

10.7 Top 3 Risks & Mitigations

Risk 1: Dataset date-range mismatch

The climate dataset is from 2013 to 2019 which shows missing data for the COVID-19 period and only has overlap data for. This will only enable the inclusion of any climate–price analysis for the period 2013-2019. Mitigation: make it clear in the report and dashboard if an analysis has been performed for a climate range (2013-2019) and make it clear if the analysis was only a price analysis (not one for a climate range) and not mislead the reader regarding the period that the chart covers.

Risk 2: Geographic mismatch between market location and supply origin

The product may be available on the market, and its origin in another location. The price data does not contain information on the location where the product is sold, so the sold at Kalimati is grown by a lot of supplying districts, meaning that a climate joint that does not include the district-average would be a naive approach to the actual climate driver. Before implementing any of the climate features, think of what the documented supplying districts are for Kalimati - not an assumption, but a data-preparation step, where the supplying districts are verified from published sources from the market authorities.

Risk 3: Technical complexity of advanced Tableau visuals under time pressure

In a time-sensitive situation, some of these types of charts may be challenging to add to Tableau, or it could be that these types of charts are a constraint at the end of the project. Mitigation: Make and finish charts easier first (bar, line, heatmap chart); use a reasonable amount of time on a more technically challenging chart; and have a simpler chart design plan for charts that can't be done within the time limit.

11. Conclusion & Future Work

11.1 Research and Dashboard Contribution

This project started with the clear aim of filling a particular and well-documented gap identified: a price series for commodity prices that was related to an interactive exploration of daily weather observations. The dashboard met the need head-on, with a vetted data-integration pipeline and a set of chart choices specific to named stakeholder decision and not just variety for variety's sake. The dashboard is not fully interactive side to side as designed and is not fully addressing the colour access issue as designed and it does not address the Anscombe demonstration as designed. All these areas are explained at the same level of evidence as the strengths of the dashboard and are not a criterion of the Outstanding standard.

11.2 Practical Value

The five recommendations are associated with individual charts and decisions by the KFVMDBs and by the Department of Agriculture as opposed to being general suggestions about dashboard adoption. Their common ground is that the diesel dashboard is a useful tool for capturing the situation at the commodity and district level which is not otherwise evident in the current public information system, like the single-aggregated figure from Nepal Rastra Bank, reproduced in the national news media.

11.3 Limitations Carried Forward

As described above there are three limitations that have been identified that make the dashboard less useful in the short-term:

- the limitation of the data range (2013-2019);
- the ability of the Year filter to be applied to only one worksheet;
- the un-resolved categorical colour palette.

None of these are architectural problems, they are all problems that have boundaries, they are all specific problems, and they are all good enough for any team that might be doing the same work.

11.4 Future Enhancements

The most immediate improvement of the underlying climate dataset is to keep it going beyond 2019 so that the dashboard can point to an event that inspired the background research that formed this project, and not to the previous analogue. A real-time or near-real-time data connection, instead of the present static extract-based build, would enable the dashboard to be used as an operational monitoring tool instead of one that is only used for analysis and reporting in the past; this would require the KFVMDB's own price-index publication process to support a compatible feed, which is beyond the scope of this project. On the other hand, the simplest version of a weather forecast model, a simple regression model, can be incorporated into the dashboard to transform it from descriptive/diagnostic to anticipatory. The map view would allow a more specific decision to be made based on data from the supply chain other than volume of rainfall, such as the status of transport routes, or documented district to market travel times. Finally, this build is not optimized for a mobile user; it has been designed and tested on the desktop and is being used by the KFVMDB field user who is logging on to the dashboard from a mobile device and will be using the original automatic layout, rather than a specially designed one.

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13. Appendices

Appendix A — Contribution Log

Role	Member	Actual Work
Team Leader	Sapana Tamang	Built the dashboard, KPI summary & the chart layout.
Research Domain Lead	Anish Bhandari	Sourced the datasets, Took lead of literature review and the research & visualization gap. Helped with the reporting.
Data Engineer	Osin Dhamala	Pulled different strands: merging data, dashboard & the report. Included the DV principles in all of them.
Visualization Designer	Osin Dhamala	Built the varieties of the charts as per the project’s requirements & worked on Tableau.
Documentation & Presentation	Karin Maharjan	Helped with the report structure, table of contents and

		the self-reflection contributions.
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Appendix B — Peer Evaluation

Submit confidentially via TImeS. Rate each member 1 (poor) to 5 (excellent).

Criterion	M1 Sapana Tamang	M2 Anish Bhandari	M3 Osin Dhamala	M4 Karin Maharjan
Technical Contribution	3	3	4	3
Research & Inquiry (TGC 4)	3	4	4	3
Reliability & Attendance	5	5	5	5
Collaboration & Communication	5	5	5	5
Overall	4	4	4	4

Appendix C — AI Tool Declaration

Generative AI tools were used during this project in the following capacities, disclosed here in the interest of the originality and integrity standard discussed in Section 13:

- to assist with drafting sections of this report from evidence supplied by the team (Tableau workbook files and the Python preprocessing notebook);
- to provide critical review of dashboard design decisions against visualization theory and the module's marking rubric; and to assist with debugging and structuring specific Tableau dashboard actions.
- All data analysis, the Python preprocessing pipeline was frame worked
- And all dashboard design decisions documented in Section 11 were carried out and authored by the team;

AI-assisted drafting was reviewed and edited by the team prior to submission. Team members should confirm this declaration is accurate and complete before submission and amend it directly if any detail here does not match actual practice.

Github

Link:

https://github.com/SapanaTamang/Data_Visualization_Dashboard/blob/main/README.md

